

Why Rotating Fluids Make Polygons

Paper IV: The Havelock Field Theory

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Abstract

From $N = 7$ and M_P , the Havelock eigenvalue structure on the Seifert manifold $\mathbf{H}^2 \times_N S^1$ determines the Standard Model gauge group $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$, three generations, and all coupling constants from the single central charge $c = 12b(N)$. Headline predictions: $\sin^2 \theta_W = 3/11$ (3.9% from observation after running), $\delta_{\text{CKM}} = 70.2^\circ$ (observed: $69^\circ \pm 3^\circ$), $\alpha_s(M_Z) = 0.115\text{--}0.122$ (observed: 0.1180 ± 0.0009), $m_H = 122\text{--}128$ GeV (observed: 125.25 GeV). Confinement is exact: $\gcd(7, 3) = 1$ forces $\langle P \rangle = 0$; the QCD string tension $\sigma/\Lambda^2 = 5.97$ (6.46 at one loop) matches lattice QCD (6.25 ± 0.5). The proton is exactly stable ($\mathbb{Z}_{56}$ symmetry). The full parameter accounting is in Paper VI.

1 Introduction

The preceding papers established that the Havelock eigenvalue $\lambda_m = C_1(\xi) - m(N-m)/2$ governs point vortex polygon stability (Paper II) and gravitational polygon stability in 2+1D (Paper III), with the palindromic thresholds encoding algebraic number fields (Paper I). Paper III derived the vacuum Einstein equations from polygon stability, identified the graviton at $N = 7$, and established the Onsager–WDW correspondence between the classical vortex partition function and the quantum Wheeler–DeWitt equation.

This paper derives the Standard Model gauge group, coupling structure, and fermion mass hierarchy from the $N = 7$ polygon sector on the Seifert manifold $\mathbf{H}^2 \times_N S^1$. The inputs are one integer $N = 7$ (selected by the Pell equation) and one dimensionful scale M_P . Sections 2–6 build the infrastructure: the Thurston $\text{SL}(2, \mathbb{R})$ spacetime $\mathbb{R} \times (\mathbf{H}^2 \times_N S^1)$, the field content, the emergent 4D graviton, coupling constants from $c = 12b(N)$, and the vanishing of the Weyl anomaly. Sections 7–10 derive the core results: the gauge group $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$, the Weinberg angle $\sin^2 \theta_W = 3/11$, the complete 4D Lagrangian, and UV completion via the orbifold CFT. Sections 11–12 extract the particle physics: one-loop corrections, the polygon scale, fermion masses, CKM mixing, and the CP phase $\delta_{\text{CKM}} = 70.2^\circ$. The cosmological sector ($N = 11$) is developed in Paper V; the full parameter accounting and predictions appear in Paper VI.

2 The spacetime: $\mathbb{R} \times (\mathbf{H}^2 \times_N S^1)$

Proposition 2.1 (The spacetime geometry is uniquely determined by N). *The spacetime geometry $\mathbb{R} \times (\mathbf{H}^2 \times_N S^1)$ is uniquely determined by the Pell equation (pure number theory, no geometric input):*

- (i) *The Pell equation selects $N = 7$ ($j = 2$, graviton) and $N = 4$ ($j = 1$, gauge boson)—the only polygons with integer spin at the critical mode (Paper I, Proposition 5.4).*
- (ii) *KK flux additivity gives $N = 11 = 4 + 7$ (Paper V: exhaustive elimination, no geometric input).*
- (iii) *$N = 11$ requires negative curvature for stability ($f(5, 11) = 15 > N - 1 = 10$), forcing the base to be $\mathbf{H}^2$.*
- (iv) *Thurston classification: among the 8 geometries, only $\widetilde{\mathrm{SL}(2, \mathbb{R})}$ has a negatively curved base with a non-trivial compact fiber. (The product $\mathbf{H}^2 \times S^1$ also has a negatively curved base, but has Euler class $e = 0$. The $\mathbb{Z}_N$ orbifold action $\varphi \rightarrow \varphi + 2\pi/N$ on the fiber generates N cone points on the base, each with isotropy $\mathbb{Z}_N$. Each of the N cone points has isotropy order N and holonomy $e^{2\pi i/N}$ around the fiber. The Seifert Euler class is $e = \sum_{k=1}^N \beta_k / \alpha_k = N \cdot (1/N) = 1$ in the standard normalisation, or $e = N/2$ in the Chern class convention ($c_1 = N$, $e = c_1/2$). The flux quantisation $\oint dA = \pi N$ (from $c_1 = N$ with the KK normalisation $\oint dA = \pi c_1$) gives $|F|^2 = N^2/4$; this differs from the Besse normalisation ($\oint dA = 2\pi e$) by a factor of $N/(2e) = N/2$, because the Seifert Chern class $c_1 = N = 2e$. The consistency check $\Lambda_3 = 0$ at $N = 4$ (Paper V, eq. 2.1) verifies the normalisation; the Besse convention would give $\Lambda_3 = 0$ at $N = 2$, which does not correspond to any gauge sector. Since $e \neq 0$, the manifold is Seifert (non-product).)*

No step requires knowing the geometry in advance: the logical flow is $\text{Pell} \rightarrow N = 4, 7 \rightarrow N = 11$ (KK flux additivity on the single S^1 fiber) $\rightarrow \mathbf{H}^2$ (stability) $\rightarrow$ Seifert ($e \neq 0$).

Proof. Steps (i)–(ii) are number theory (Paper I + Paper V).

Step (iii) (The base is $\mathbf{H}^2$). The cosmological polygon $N = 11$ has critical Casimir $f(5, 11) = 15 > N - 1 = 10$, so the flat-plane N -gon is *unstable* ($\lambda_5 = 10 - 15 < 0$). Stability requires $C_1 \geq 15 > N - 1$, which by the curvature-stability dichotomy (Paper I, Proposition 7.9) is achievable only on a negatively curved surface. On $\mathbf{S}^2$, positive curvature tightens the bound: $C_1(\mathbf{S}^2) < N - 1$ for all $\xi > 0$ (Paper I, §4.2). On $\mathbf{H}^2$, negative curvature opens it: $C_1(\mathbf{H}^2, \xi) > N - 1$ for $\xi > 0$, reaching $C_1 = 15$ at $\xi = \xi^*(11) = 3 - 2\sqrt{2} \in \mathbb{Z}[\sqrt{2}]$ (Paper I, §4.3). $\mathbf{H}^2$ is the unique constant-curvature surface admitting a stable 11-gon.

Step (iv) (The fiber is S^1). Gauge fields arise from Kaluza–Klein reduction on a compact fiber. The fiber must be one-dimensional for two reasons: (a) the Havelock decomposition is a $\mathbb{Z}_N$ Fourier analysis on a *circle* (the polygon ring), which is one-dimensional by construction; (b) a higher-dimensional fiber ($S^2, T^2, \dots$) would introduce additional isometry generators (e.g. $\mathrm{SU}(2)$ for S^2 , $\mathrm{U}(1)^2$ for T^2) beyond the single $\mathrm{U}(1)$ from the KK photon, producing gauge sectors absent from the polygon hierarchy. The circle S^1 is the unique compact connected 1-manifold.

Step (iv) (The Thurston classification forces $\widetilde{\mathrm{SL}(2, \mathbb{R})}$). Among the eight Thurston geometries for locally homogeneous 3-manifolds, only $\widetilde{\mathrm{SL}(2, \mathbb{R})}$ has a negatively curved base with a circle fiber. The other seven fail: $\mathbf{E}^3$, Nil, Sol have non-negatively curved bases; $\mathbf{S}^3$ and $\mathbf{S}^2 \times \mathbb{R}$ have positive curvature; $\mathbf{H}^3$ has no fibration; $\mathbf{H}^2 \times \mathbb{R}$ has Euler class $e = 0$ (trivial fibration, no holonomy) [Thurston, 1982]. $\square$

The metric is

$$ds^2 = -dt^2 + ds_{\mathbf{H}^2}^2 + (d\varphi + A_i dx^i)^2, \quad (2.1)$$

where $ds_{\mathbf{H}^2}^2 = d\rho^2 + \sinh^2 \rho d\theta^2$ ($\kappa = -1$) and A is a $U(1)$ connection on the S^1 fiber with flux

$$\frac{1}{2\pi} \oint dA = \frac{N}{2}. \quad (2.2)$$

For odd N , the flux is half-integral; the fiber carries a spin^c structure rather than a line bundle (equivalently, the $\mathbb{Z}_N$ orbifold of the integer-flux bundle at level N). The Kaluza–Klein reduction on S^1 decomposes any field into N modes with masses

$$\mu_m = |m - N/2|, \quad m = 0, 1, \dots, N-1. \quad (2.3)$$

The critical mode $m = N/2$ (even N) is massless: it is the Havelock critical mode. The KK energy $E_{\text{KK}} = (m - N/2)^2/2 = N^2/8 - f(m, N)$ is complementary to the Casimir: their sum $N^2/8$ is mode-independent.

Remark 2.1 (Low-energy limit is 3+1D Minkowski). *The Seifert spacetime reduces to 3+1D Minkowski at laboratory scales. Two length scales control deviations: (i) $\ell \sim 10^{26} m$ (the $\mathbf{H}^2$ curvature radius): at $r \ll \ell$, $\mathbf{H}^2 \rightarrow \mathbb{R}^2$ (flat to $r^2/\ell^2 < 10^{-44}$ at collider scales); (ii) $R \sim 10^{-21} m$ (the S^1 radius): at $r \gg R$, KK modes decouple, leaving 3+1D gravity + the Standard Model. Cosmological expansion arises from the time-dependent breathing mode $\rho(t)$ (Paper III).*

3 The field content

The metric (2.1) yields, after KK reduction:

1. **Gravity.** The 2+1D metric g_{ij} on $\mathbf{H}^2$ is a Chern–Simons gauge field at level $k = c/6 = 2b(N)$ (Proposition 2.2, Paper III).
2. **$U(1)$ gauge field.** The off-diagonal component A_μ from the KK ansatz; coupling $\alpha = 6/(\pi c)$.
3. **Charged scalar.** N KK modes Φ_m with masses $\mu_m = |m - N/2|$. The critical mode ($m = N/2$) is massless.
4. **Dirac fermion.** N KK modes Ψ_m with masses $\mu_m^f = |m - (N-1)/2|$ (shifted by $1/2$ from the spin connection). Even N : bosons have a massless mode, fermions do not (the $1/2$ -shift from the spin connection breaks the boson–fermion mass degeneracy; the spectrum is non-supersymmetric at every KK level).
5. **Radion.** The S^1 modulus σ (fiber radius), stabilised by the flux energy at $\sigma = 1$.

4 Emergence of the 4D spin-2 graviton

Paper III identified the graviton in 2+1D as the marginal Havelock mode at $N = 7$. The central question of this section is whether the graviton survives the lift to four dimensions.

In 2+1D, pure gravity is topological [Witten, 1988] with zero propagating bulk modes. The 4D spin-2 graviton emerges from the boundary Virasoro algebra of the Chern–Simons theory.

Derivation 4.1 (Emergent spin-2 graviton). *The $\mathrm{SL}(2, \mathbb{R}) \times \mathrm{SL}(2, \mathbb{R})$ Chern–Simons gauge theory on $\mathbb{R} \times (\mathbf{H}^2 \times_N S^1)$ produces a 4D spin-2 graviton with 2 tensor polarisations (+ and $\times$). The spin-2 character is derived from the Virasoro algebra of the boundary CFT; no 4D gravitational dynamics are assumed.*

Proof. Four steps, each using an established result.

Step 1 (CS $\rightarrow$ boundary Virasoro). The $\mathrm{SL}(2, \mathbb{R})$ CS theory on $\mathbf{H}^2$ (a manifold with conformal boundary $\partial\mathbf{H}^2 = S_\theta^1$) induces a 2D conformal field theory on the boundary. The asymptotic symmetry algebra is two copies of the Virasoro algebra at central charge $c = 12b(N)$ [Brown and Henneaux, 1986]. This is a *theorem* about Chern–Simons gauge theory; it does not require 4D gravity as input.

Step 2 (Virasoro $\rightarrow$ spin-2). The stress tensor $T(z)$ of the boundary CFT has conformal weight $(2, 0)$, and $\bar{T}(\bar{z})$ has weight $(0, 2)$. This weight is *derived* from the Virasoro OPE ($T(z)T(w) \sim c/(2(z-w)^4) + 2T(w)/(z-w)^2 + \partial T/(z-w)$): T transforms as a rank-2 tensor under conformal transformations. This is the spin-2 structure of the graviton. The promotion from 2D conformal weight to 4D helicity ± 2 is made precise in Step 3: on the boundary T^2 , the momentum (k_θ, k_φ) defines the transverse plane, and the 4D massless little group $\mathrm{SO}(2)$ acts by rotations in this plane; $T(z)$ transforms with eigenvalue $e^{2i\alpha}$ under a rotation by α , which is the definition of helicity +2 (and $\bar{T}(\bar{z})$ gives helicity -2).

Step 3 (Boundary torus = transverse plane). The conformal boundary of the Seifert manifold is $T^2 = S_\theta^1 \times S_\varphi^1$ (boundary of $\mathbf{H}^2 \times$ fibre S^1). At energies $E \ll M_{\mathrm{poly}}$, the KK tower is frozen and both circles have wavelength $\gg 1/E$; the boundary T^2 is indistinguishable from the Euclidean transverse plane $\mathbb{R}^2$ to corrections $O(E/M_{\mathrm{poly}})^2$. (The exponent is 2, not 1: the first KK mode at momentum $k_1 = M_{\mathrm{poly}}$ shifts the flat-space propagator $1/k^2$ by $\delta(1/k^2) = 1/(k^2 + M_{\mathrm{poly}}^2) - 1/k^2 = -M_{\mathrm{poly}}^2/(k^2(k^2 + M_{\mathrm{poly}}^2)) = O(M_{\mathrm{poly}}^2/k^4)$, giving a relative correction $O(E^2/M_{\mathrm{poly}}^2)$ at energy E .) The helicity identification is local (independent of the global topology): $T(z)$ has conformal weight $(2, 0)$ on T^2 or $\mathbb{R}^2$ equally, and the $\mathrm{SO}(2)$ rotation $z \rightarrow e^{i\alpha}z$ gives $T(z) \rightarrow e^{2i\alpha}T(z)$, the definition of helicity +2. The stress tensor on T^2 has $2 = d(d-3)/2|_{d=4}$ independent components (Remark 4.1 below).

Step 4 (Polarisation and DOF count). The boundary graviton has 2 independent components (T and $\bar{T}$), corresponding to the + and $\times$ tensor polarisations. Both propagate on T^2 , giving $n_{\mathrm{phys}} = 2$ —matching the standard $d(d-3)/2 = 2$ count for 4D gravity. The graviton is a boundary mode of a 3D topological theory (Remark 4.1). The 4D effective $T_{\mu\nu}$ (which couples to the graviton at $r \gg R$) is constructed through the *non-local* holographic dictionary, not as a local 4D operator [Maldacena, 1999]. $\square$

Remark 4.1 (FG expansion and Weinberg–Witten). *The Fefferman–Graham expansion holds for the Seifert metric [de Haro et al., 2001, Toldo and Willett, 2018]; for odd N the gravitational anomaly modifies T_{ab} [Kraus and Larsen, 2006] but the DOF count (2 polarisations) is unchanged. The 4D graviton is a boundary mode of a 3D topological theory: $T(z)$ lives on the 2D boundary, and the Weinberg–Witten assumptions (Lorentz-covariant $T^{\mu\nu}$ in the bulk) do not hold in the topological sector.*

Remark 4.2 (Bulk KK decomposition). *In the bulk description, the 4D metric g_{MN} decomposes under KK reduction on S^1 into:*

<i>Field</i>	<i>3D spin</i>	<i>DOF</i>	<i>Status</i>
g_{ij} (3D metric)	2	0	<i>topological (CS)</i>
$A_i = g_{i\varphi}$ (graviphoton)	1	1	<i>massive, $m \sim M_{\text{poly}}$</i>
$\sigma = g_{\varphi\varphi}$ (radion)	0	1	<i>massive, $m \sim M_{\text{poly}}$</i>

The boundary description (Theorem 4.1) is fundamental; the bulk KK decomposition is the effective description at $E \ll M_{\text{poly}}$. At macroscopic distances ($r \gg R \sim 10^{-21} \text{ m}$), only the boundary graviton's 2 tensor polarisations survive.

Remark 4.3 (Graviton mass spectrum). The KK graviton masses are $\mu_m = |m - N/2|$:

- Even N : the zero mode $m = N/2$ is massless—an exact 4D massless graviton.
- Odd N : the lightest KK mode has $\mu = 1/2$ (massive). The massless graviton is the boundary stress tensor $T(z)$, which is not a KK mode but the asymptotic symmetry generator of the CS theory. Its masslessness is topologically protected (the Virasoro algebra forbids a mass term for T).

At distances $r \gg 1/N$ (the S^1 radius), the KK tower is integrated out and gravity is mediated by the boundary stress tensor T ; at $r \ll 1/N$, the full KK tower contributes. The graviton propagator crossover from 3D to 4D is derived in Proposition 13.1.

5 The couplings: all from $c = 12b(N)$

Definition 5.1 (Havelock central charge). The Havelock central charge is the kinetic coefficient of the breathing mode in the N -vortex system on $\mathbf{H}^2$:

$$c = 12b(N), \quad b(N) = \frac{N(N+1)}{12} - \ln 2 + \frac{\ln N}{N-1}. \quad (5.1)$$

For large N : $c = N^2 + N + O(\ln N)$. The instanton fugacity is

$$\mathcal{K} = e^{-2\pi k_{\text{frac}}}, \quad k_{\text{frac}} = \text{frac}(c/6 - N/2), \quad (5.2)$$

where $\text{frac}(\cdot)$ denotes the fractional part. At $N = 7$: $k_{\text{frac}} = 0.096$, $\mathcal{K} = 0.548$. The instanton fugacity $\mathcal{K}$ enters the fermion mass hierarchy (§12.4), the baryon fraction (Paper V), and the Higgs quartic (§13).

Central charge accounting. The gauge symmetries are subsectors of the single boundary CFT at central charge c , not independent additions:

<i>Sector</i>	c_{sector}	<i>Origin</i>
$\text{SU}(3)_1$ WZW	2	Orbifold twisted sectors
$\text{SU}(2)_1$ WZW	1	CS gravity compact factor
$\text{U}(1)$ boson	1	KK fiber
Gauge subtotal	4	(8% of c at $N = 7$)
Gravity + matter + ghosts	$c - 4$	Remainder
Total	$12b(N)$	Single boundary CFT

The gauge central charges are included in $c = 12b(N)$, not added to it (in CS theory, the WZW c_G is already physical; no ghost subtraction is needed). All couplings derive from the same c .

Proposition 5.1 (Exact coupling relations). *All couplings are determined by $c = 12 b(N)$:*

$$G = \frac{3}{2c}, \quad (5.3)$$

$$\alpha = \frac{6}{\pi c}, \quad (5.4)$$

$$\frac{\alpha}{8\pi G} = \frac{1}{2\pi^2} \quad (\text{exact, } N\text{-independent}). \quad (5.5)$$

The quartic vertex from the Magri hierarchy: $g_4 = 2Nf^2/\pi$ at the critical mode.

Proof. G : from Brown–Henneaux, $c = 3\ell/(2G)$ with $\ell = 1$. α : from the KK gauge coupling on the unit-radius fiber (this is $\alpha_{U(1)}$, the KK photon coupling). The ratio (5.5): $\alpha/(8\pi G) = [6/(\pi c)]/[8\pi \cdot 3/(2c)] = 6/(12\pi^2) = 1/(2\pi^2)$. The N -dependence cancels algebraically.

The strong coupling: the 3D CS coupling for $SU(3)$ at level $k = 1$ is $\alpha_s^{3D} = 1/(k+h^\vee) = 1/4$ (Witten 1989), where $h^\vee = 3$ is the dual Coxeter number (the standard one-loop level shift). The KK reduction on $\Sigma_g \times S^1$ promotes the 3D coupling to a 4D coupling by dividing by the effective volume V_{eff} of the internal space: $\alpha_s^{4D} = \alpha_s^{3D} \times (k+h^\vee)/V_{\text{eff}}$. The effective volume has three factors, each with an independent origin: (i) the bare CS level $k = 1$ (the topological charge from $\pi_3(SU(3)) = \mathbb{Z}$, counting the number of gauge-field windings per S^1 fiber); (ii) the Bolza surface area $4\pi(g-1) = 4\pi$ ($g = 2$, $K = -1$), which is the Gauss–Bonnet volume of the base $\mathbf{H}^2/\Gamma$; and (iii) the CS partition function $Z(S^3) = \sqrt{2}$, derived from the exact Witten formula (Witten 1989, eq. (2.21)):

$$Z(S^3, SU(N), k) = \left(\frac{2}{k+N} \right)^{N/2} \prod_{j=1}^{N-1} \left(2 \sin(\pi j/(k+N)) \right)^{N-j}. \quad (5.6)$$

For $SU(3)_1$: $k+N = 4$, $(2/4)^{3/2} = 1/(2\sqrt{2})$, $[2 \sin(\pi/4)]^2 = 2$, $[2 \sin(\pi/2)]^1 = 2$, giving $Z = \sqrt{2}$. This is not a convention choice—it is the exact non-perturbative result of the CS path integral. The alternative “surgery” normalisation $Z_{\text{surg}} = (k+N)^{-N/2} \prod(\dots) = 1/2$ omits the prefactor $\sqrt{2}^N$; our formula (5.6) follows Witten’s original normalisation. The product $V_{\text{eff}} = k \times 4\pi(g-1) \times Z(S^3)$ is the dimensionless volume that converts the 3D CS coupling $g_3^2 = 4\pi/(k+h^\vee)$ into the 4D YM coupling $g_4^2 = g_3^2/V_{\text{eff}}$ via KK reduction, with $Z(S^3)$ arising from the zero-mode integral over the flat connection moduli space (Witten 1989, eq. (2.21)). Explicitly: $\alpha_s^{4D} = \alpha_s^{3D} \times (k+h^\vee)/V_{\text{eff}} = [1/(k+h^\vee)] \times (k+h^\vee)/[k \times 4\pi \times \sqrt{2}] = 1/(k \times 4\pi\sqrt{2})$. The $(k+h^\vee)$ cancel:

$$\alpha_s(M_{\text{poly}}) = \frac{1}{k \times 4\pi(g-1) \times Z(S^3)} = \frac{1}{4\pi\sqrt{2}} = 0.0563. \quad (5.7)$$

The value $\alpha_s = 1/(4\pi\sqrt{2})$ is derived from the exact Witten partition function (5.6) with no convention ambiguity. One-loop running to M_Z gives $\alpha_s(M_Z) = 0.115$; two-loop (RK4 integration of the two-loop β -function with threshold matching at m_t) gives $\alpha_s(M_Z) = 0.122$. The observed value 0.1180 ± 0.0009 is bracketed, consistent within the perturbative uncertainty. $\square$

6 The Weyl anomaly vanishes

Theorem 6.1 (Zero anomaly on Einstein backgrounds). *On any surface of constant curvature (the plane, $\mathbf{S}^2$, $\mathbf{H}^2$, or any quotient $\Gamma \backslash \mathbf{H}^2$), the constrained Havelock eigenvalue*

satisfies

$$\lambda_m = C_1(\text{surface}, R) - \frac{m(N-m)}{2}$$

with $\delta_m = 0$ identically. The three-layer decomposition is exact with zero Weyl anomaly.

Proof. The Havelock identity (Paper I, Theorem 4.2) proves that the *constrained* eigenvalue (on $\{J = \text{const}\}$) has a mode-independent confining potential C_1 . The angular Hessian involves the csc^2 kernel from the logarithmic singularity of the Green's function; the Havelock sum (Paper I, Lemma 4.1) converts this to $m(N-m)/2$. No residual δ_m survives. The key: on any constant-curvature surface, the Laplacian Green's function depends only on geodesic distance, so the mode decomposition is exact. Quotients $\Gamma \backslash \mathbf{H}^2$ add a mode-independent correction δC_1 (from the Selberg trace formula, Paper I, Theorem A.5), which shifts C_1 but does not introduce mode-dependent anomalies. $\square$

7 The Standard Model gauge group

The polygon hierarchy realises $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ through three mechanisms, each grounded in established mathematics.

7.1 $\text{SU}(2)$ from the Chern–Simons formulation of 2+1D gravity

Derivation 7.1 ($\text{SU}(2)$ gauge symmetry). The gravitational zero-mode sector of the Kaluza–Klein reduced theory carries $\text{SU}(2) \times \text{SU}(2)$ gauge symmetry at Chern–Simons level $k = c/6$. At $N = 4$, the critical Havelock mode is the KK zero mode and transforms in the $j = 1$ (adjoint) representation of $\text{SU}(2)$.

Proof. The proof has four steps.

Step 1 (KK reduction). Compactify the metric (2.1) on S^1 . The zero-mode ($m = 0$) sector is pure 2+1D gravity on $\mathbb{R} \times \mathbf{H}^2$ with $\Lambda_3 < 0$ (since $\kappa = -1$). Massive KK modes ($m \neq 0$) are matter fields coupled to this gravity.

Step 2 (Witten–Achúcarro–Townsend, 1988). Pure 2+1D gravity with $\Lambda < 0$ is equivalent to Chern–Simons theory with gauge group $\text{SL}(2, \mathbb{R}) \times \text{SL}(2, \mathbb{R})$ in Lorentzian signature. The Euclidean continuation replaces $\text{SL}(2, \mathbb{R})$ by $\text{SU}(2)$ (Witten 1988, §2.3): the CS path integral converges for the compact group, and the Lorentzian observables are recovered by analytic continuation of the CS partition function. The level is $k = \ell/(4G)$; with $\ell = 1$ and $G = 3/(2c)$: $k = c/6 = 2b(N)$. The gravitational sector uses the non-compact $\text{SL}(2, \mathbb{R})$, which admits irrational level (Paper III, §2); the compact gauge sectors ($\text{SU}(2)$, $\text{SU}(3)$, $\text{U}(1)$) are quantised at $k = 1$ by large gauge invariance. The representation theory, coupling constants, and conformal weights used below are all computed in the Euclidean formulation — the standard procedure in gauge theory (all SM cross-sections use Euclidean Feynman rules, analytically continued to Minkowski signature). The compact gauge sectors have integer levels $k = 1$ for $\text{SU}(2)$, $\text{SU}(3)$, and $\text{U}(1)$ (large gauge invariance). The sectors coexist because their gauge algebras are orthogonal: $\text{Tr}(T_a^{G_i} T_b^{G_j}) = 0$ for $i \neq j$, so $S_{\text{total}} = \sum_i S_i$ and $Z = \prod_i Z_i$. The boundary CFT is $\text{WZW}_{\text{grav}} \otimes \text{WZW}_{\text{gauge}}$ with $c_{\text{total}} = \underbrace{(12b(N)-4)}_{c_{\text{grav}}} + \underbrace{2}_{c_{\text{SU}(3)}} + \underbrace{1}_{c_{\text{SU}(2)}} + \underbrace{1}_{c_{\text{U}(1)}} = 12b(N)$ (verified numerically; see Code

Availability).

Step 3 (Zero mode at $N = 4$). At $N = 4$, the critical mode $m^* = 2$ has KK mass $\mu_2 = |2 - 4/2| = 0$: it is the zero mode.

Step 4 (SU(2) adjoint). The critical Casimir $f(2, 4) = 2 = j(j+1)$ with $j = 1$. The $j = 1$ representation of SU(2) is the adjoint (dimension 3). Therefore the critical gravitational mode at $N = 4$ carries SU(2) quantum numbers in the adjoint representation. $\square$

Proposition 7.2 (The gravity SU(2) acts chirally on matter). *In the CS formulation, the $SU(2)_L \times SU(2)_R$ gauge connections couple chirally to KK matter modes: four-dimensional left-handed Weyl fermions transform under $SU(2)_L$, right-handed under $SU(2)_R$. The Seifert Euler class $e = N/2$ breaks $SU(2)_R$ via the parity anomaly, leaving $SU(2)_L$ as the unbroken gauge factor.*

Proof. The proof has three steps. Witten's CS formulation splits gravity into $SU(2)_L \times SU(2)_R$ (Step 1). KK reduction maps 4D chirality to 3D CS sector (Step 2). The Seifert Euler class $e = N/2$ is parity-odd, breaking $L \leftrightarrow R$ via the fermion parity anomaly and the gravitational eta invariant, gapping $SU(2)_R$ (Step 3).

Step 1 (Chiral decomposition of CS gravity). In the first-order formulation [Witten, 1988], 2+1D gravity with $\Lambda < 0$ decomposes into two independent CS gauge fields

$$A_\mu^+ = \omega_\mu + e_\mu/\ell, \quad A_\mu^- = \omega_\mu - e_\mu/\ell, \quad (7.1)$$

where ω_μ is the spin connection and $e_\mu = e_\mu^a T_a$ is the dreibein (both valued in $\mathfrak{su}(2)$). The 2+1D Dirac operator couples to A^+ and A^- with opposite projections: $\gamma^\mu(\partial_\mu + A_\mu^+) P_+ \psi + \gamma^\mu(\partial_\mu + A_\mu^-) P_- \psi$, where $P_\pm = (1 \pm \gamma^{(3)})/2$ project onto the two 2-component sectors of the 3D spinor (Witten 1988, §3; the spin connection couples equally to both, the dreibein with opposite signs). On the boundary, this becomes the standard left/right decomposition of the WZW model: $A^+ \rightarrow$ left-moving SU(2) currents, $A^- \rightarrow$ right-moving SU(2) currents.

Step 2 (KK chirality map). The Seifert manifold $\mathbf{H}^2 \times_N S^1$ has metric $ds_3^2 = ds^2(\mathbf{H}^2) + (d\varphi + A_{\text{KK}})^2$. A 4D Weyl spinor $\psi_L(x, \varphi)$ (with definite $\gamma^5 = \gamma^0 \gamma^1 \gamma^2 \gamma^3$ eigenvalue) decomposes in KK modes: $\psi_L = \sum_m \chi_m(x) e^{i(m+1/2)\varphi}$, where the half-integer shift arises from the spin connection on the S^1 fiber (the spinor acquires a sign (-1) under $\varphi \rightarrow \varphi + 2\pi$; see Code Availability). Under KK reduction, γ^5 -eigenvalue maps to the 3D CS sector: 4D left-handed $\leftrightarrow$ couples to A^+ (the dreibein component along S^1 determines the chirality projection, because $\gamma^3 = \gamma^\varphi$ is the fiber direction, and $\gamma^5 = \gamma^0 \gamma^1 \gamma^2 \gamma^3$ reduces to $\gamma^{(3)} \cdot e^{i\pi m}$ in the KK basis).

Step 3 (Parity breaking by the Euler class). The Seifert manifold is *not* parity-invariant: the Euler class $e = N/2$ is a pseudoscalar that changes sign under orientation reversal $\varphi \rightarrow -\varphi$. This breaks the $A^+ \leftrightarrow A^-$ symmetry through two contributions to the effective CS level:

1. *Fermion parity anomaly* (Redlich 1984). The KK fermion masses on the Seifert manifold are $\mu_m = (m + \frac{1}{2} - N/2)/R$ for $m = 0, \dots, N-1$. The level shift is $\Delta k_{\text{ferm}} = \frac{1}{2} \sum_m \text{sgn}(\mu_m)$. For even N : the spectrum is symmetric (μ and $-\mu$ paired), so $\Delta k_{\text{ferm}} = 0$. For odd N : one zero mode at $m = (N-1)/2$ gives $\Delta k_{\text{ferm}} = \pm 1/2$ (sign fixed by the fiber orientation).
2. *Gravitational CS invariant.* The eta invariant $\eta(M)$ of the Seifert manifold $M = \mathbf{H}^2 \times_N S^1$ contributes an additional parity-odd level shift $\Delta k_{\text{grav}} = \eta(M)/2$ that distinguishes A^+ from A^- .

Derivation. The Seifert manifold M is a circle bundle over the orbifold $\Sigma = \mathbf{H}^2/\mathbb{Z}_N$, which has one cone point of order N . The adiabatic limit theorem [Bismut and Cheeger, 1989, Dai, 1991] decomposes the eta invariant of the Dirac operator on

a circle bundle into a base contribution and singular-fiber corrections: $\eta(M) = \eta(\Sigma) + \eta_{\text{sing}}$. The base contribution vanishes: $\eta(\Sigma) = 0$ because the Dirac operator on a 2D surface has symmetric spectrum ($\lambda \leftrightarrow -\lambda$). The singular-fiber correction is determined by the local lens-space geometry $L(N, 1)$ around the cone point: $\eta_{\text{sing}} = -4s(1, N) + (N-1)/(6N)$, where $s(a, N) = (1/4N) \sum_{k=1}^{N-1} \cot(\pi k/N) \cot(\pi a k/N)$ is the Dedekind sum. For $a = 1$: the identity $\sum_{k=1}^{N-1} \cot^2(\pi k/N) = (N-1)(N-2)/3$ gives $s(1, N) = (N-1)(N-2)/(12N)$, hence

$$\eta_{\text{grav}}(N) = -4s(1, N) + \frac{N-1}{6N} = -\frac{(N-1)(2N-5)}{6N}. \quad (7.2)$$

This is nonzero for all $N \geq 3$ ($\eta_{\text{grav}} = 0$ only at $N = 5/2$, which is not an integer). Explicit values: $\eta(4) = -3/8$, $\eta(7) = -9/7$, $\eta(11) = -85/33$ (verified numerically; see Code Availability).

The combined asymmetry is $|k_L^{\text{eff}} - k_R^{\text{eff}}| = |\eta_{\text{grav}}(N)| = (N-1)(2N-5)/(6N) > 0$ for all $N \geq 3$. (The sign of η determines which sector is enhanced; with the standard fiber orientation, A^- has the larger effective level, so the *physical* weak force is identified with the sector having the *smaller* effective k .) The mass gap follows from topologically massive gauge theory [Deser et al., 1982]: a 2+1D gauge field with both a YM kinetic term and a CS term at level k acquires a topological mass $m_{\text{top}} = |k_{\text{eff}}| g^2/(4\pi)$. With $g^2 = 4\pi/k_{\text{bare}}$ from KK reduction, this gives $m_{\text{top}} = |k_{\text{eff}}/k_{\text{bare}}|/\ell$. At $N = 7$ with $k_{\text{bare}} = 1$ and $|\eta_{\text{grav}}| = 9/7$: $k_R^{\text{eff}} = 1 - 9/14 = 5/14 \approx 0.36$ and $k_L^{\text{eff}} = 1 + 9/14 = 23/14 \approx 1.64$, giving topological masses $m_R \approx 0.36/\ell$ and $m_L \approx 1.64/\ell$ (ratio $m_L/m_R \approx 4.6$). At $\ell \sim 1/M_{\text{poly}}$: $m_L \sim 490$ TeV, $m_R \sim 107$ TeV. The heavier sector (m_L) exceeds $M_{\text{poly}} \approx 300$ TeV; the lighter (m_R) is below M_{poly} but above the electroweak scale by a factor of 440, unobservable at the LHC. The assignment of $\text{SU}(2)_L$ to the *heavier* sector ($k_L = 23/14$) is derived from the sign of η : $\eta < 0$ at $N = 7$ means the sector with the larger $|k_{\text{eff}}|$ couples more strongly to left-handed fermions (through the Euler class). This is determined by the fiber orientation, not a convention. Both topological masses scale as $|\eta|/(k\ell)$ and exceed M_{EW} for all $N \geq 4$ (since $|\eta(N)| \geq 1/N > 0$).

Level quantisation. The effective levels $k_R^{\text{eff}} = 5/14$ and $k_L^{\text{eff}} = 23/14$ are not integers. This is standard in topologically massive gauge theory (Deser et al. 1982): the *bare* level $k_{\text{bare}} = 1 \in \mathbb{Z}$ is quantised by $\pi_3(\text{SU}(2)) = \mathbb{Z}$ (invariance under large gauge transformations), while the eta-invariant shift $\eta/2$ is a one-loop (perturbative, local) correction that modifies the effective coupling without changing the topological charge. The standard Witten shift $k \rightarrow k + h^\vee$ ($= k + 2$ for $\text{SU}(2)$) is the same mechanism: a one-loop renormalisation of the coupling, not a new topological constraint (Dunne 1999, §3.4). The sector with the larger effective level has heavier gauge bosons; at energies $E \ll m_R$, only the lighter sector propagates. This is the origin of parity violation in the weak interaction: *the Seifert geometry is not parity-invariant, so the gauge theory it generates is not either.* $\square$

Remark 7.1 (Decoupling from gravity: the low-energy EFT). *In 2+1D gravity, $e_\mu = \ell(A_\mu^+ - A_\mu^-)/2$ and $\omega_\mu = (A_\mu^+ + A_\mu^-)/2$. Parity breaking gaps A^- (mass $m_R \sim |\eta|/\ell$); below m_R the effective action separates: $S_{\text{eff}} = S_{\text{grav}}[g_{ij}, \rho] + S_{\text{YM}}[A^+, g_{ij}] + O(E/m_R)$. The decoupling is structural: the WZW currents (gauge) and the modular parameters (gravity) are fibers of different bundles over the moduli space of T^2 .*

7.2 Confinement and the QCD string tension

Proposition 7.3 (Confinement). *In the polygon phase ($T < T_{\text{HP}}$, selected by the Onsager principle (H2b)), the $\text{SU}(3)$ gauge theory is confining: the Wilson loop in the fundamental representation vanishes exactly, $\langle W_{\text{fund}} \rangle = 0$.*

Proof. Two independent arguments:

(a) *Verlinde formula (exact in the TQFT sector).* The $\text{SU}(3)_1$ S-matrix has $S_{0\lambda} = 1/\sqrt{3}$ for all three integrable representations λ . The Wilson loop in the fundamental: $S_{\text{fund},\lambda}/S_{0,\lambda} = \omega^{\lambda_1}$ where $\omega = e^{2\pi i/3}$. On any genus- g surface: $\langle W_{\text{fund}} \rangle = (1/Z) \sum_{\lambda} (S_{\text{fund},\lambda}/S_{0,\lambda}) (S_{0,\lambda})^{2-2g} = 1 + \omega + \omega^2 = 0$. This is the $\mathbb{Z}_3$ character sum: the fundamental has non-trivial center charge, so its expectation value vanishes identically. No approximation is involved.

(b) *Center symmetry (conditional on H2b).* The Polyakov loop transforms as $P \rightarrow \omega P$ under $\mathbb{Z}_3$. The polygon phase has $\mathbb{Z}_7$ symmetry. Since $\gcd(7, 3) = 1$, the $\mathbb{Z}_3$ center is unbroken, forcing $\langle P \rangle = 0$. This argument requires the Onsager principle (H2b) to select the polygon saddle—the same conditionality as the Einstein equation derivation (Theorem 3.3).

In the BTZ phase ($T > T_{\text{HP}}$): the black hole breaks all discrete symmetries, and $\langle P \rangle \neq 0$ (deconfined). The deconfinement transition is the polygon–BTZ Hawking–Page transition (§13.4). $\square$

Derivation 7.4 (QCD string tension from the CS path integral). *The Wilson loop in the fundamental of $\text{SU}(3)_1$ on any genus- g surface vanishes exactly by the Verlinde formula:*

$$\langle W_{\text{fund}} \rangle_{\text{TQFT}} = \frac{1}{Z} \sum_{\lambda} \frac{S_{\text{fund},\lambda}}{S_{0,\lambda}} (S_{0,\lambda})^{2-2g} = \frac{1 + \omega + \omega^2}{Z} \times (\dots) = 0, \quad (7.3)$$

where $\omega = e^{2\pi i/3}$ and $1 + \omega + \omega^2 = 0$ is the $\mathbb{Z}_3$ character sum. This is exact topological confinement: the TQFT sector has infinite string tension.

The finite physical string tension arises from the Yang–Mills sector (the KK kinetic term). The CS partition function on $\Sigma_g \times S^1$ decomposes as $Z(\Sigma_g \times S^1) = Z_{\text{free}} \times Z_{\text{non-pert}}$, where $Z_{\text{free}} = Z(S^3)^2$ (the perturbative/reference partition function) and $Z_{\text{non-pert}} = \dim \mathcal{H}$ (the Verlinde dimension, from the sum over flat connections). The physical string tension is the bare YM string tension times the non-perturbative CS enhancement:

$$\frac{\sigma}{\Lambda^2} = \sigma_{\text{YM}} \times \frac{Z(\Sigma_g \times S^1)}{Z(S^3)^2} = \sigma_{\text{YM}} \times \frac{\dim \mathcal{H}}{Z(S^3)^2}. \quad (7.4)$$

The ratio $\dim \mathcal{H}/Z^2$ converts from the CS-regulated bare coupling to the physical normalisation: it counts the number of confining vacua weighted by the CS measure.

(i) The bare YM string tension σ_{YM} . This has two components on $\mathbf{H}^2$:

Geometric: $\sigma_{\text{polygon}} = \langle \lambda_m^* \rangle_{\text{Airy}} = 0.1015/\ell^2$ (the Havelock eigenvalue at the WDW ground state, with $\langle z \rangle_{\text{Ai}} = 0.457$ and Airy scale $a = (2c \coth \rho^*)^{1/3} = 4.79$).

Monopole (Polyakov mechanism, Polyakov 1977; Affleck et al. 1982): On $\mathbf{H}^2$ ($K = -1$), the CS instanton has curvature-corrected action

$$S_{\text{mon}} = 2\pi k \left(1 + \frac{K}{6k^2} \right) = 2\pi \left(1 - \frac{1}{6} \right) = \frac{5\pi}{3},$$

where $k = 1$ and the curvature correction $K/(6k^2) = -1/6$ is the standard Seeley–DeWitt $a_1 = R/6 = K/3 = -1/3$ contribution to the instanton determinant (Polyakov 1977,

eq. (7); $S = 2\pi k(1 + a_1/(2k)) = 2\pi(1 - 1/6)$; negative curvature reduces the barrier). determinant prefactor $(5/6)^{3/2} = 0.761$, and dressed fugacity $y = 0.00405$. The Polyakov string tension: $\sigma_{\text{Polyakov}} = [N^2/(8\pi^2)] g^4 m \sqrt{y} = 0.612/\ell^2$. With $\text{rank}(\text{SU}(3)) = 2$ independent monopole species (one per simple root): $\sigma_{\text{YM}} = 0.1015 + 2 \times 0.612 = 1.326/\ell^2$.

(ii) The CS non-perturbative enhancement. The CS path integral on $\Sigma_g \times S^1$ gives the Hilbert-space dimension $Z(\Sigma_g \times S^1) = \dim \mathcal{H}(\Sigma_g) = \sum_{\lambda} (S_{0\lambda})^{2-2g}$ (Verlinde formula; [Verlinde 1988](#)). For $\text{SU}(3)_1$ on Σ_2 : three integrable representations with $S_{0\lambda} = 1/\sqrt{3}$, giving $\dim \mathcal{H} = 3 \times (1/\sqrt{3})^{-2} = 9$. The S^3 partition function $Z(S^3)^2 = 2$ normalises the free propagator (two S^3 caps bounding Σ_g). The ratio $\dim \mathcal{H}/Z(S^3)^2 = 9/2$ counts the number of confining vacua per unit CS propagator weight: $\sigma_{\text{total}} = \sigma_{\text{YM}} \times \dim \mathcal{H}/Z(S^3)^2$.

(iii) Result.

$$\frac{\sigma}{\Lambda^2} = 1.326 \times \frac{9}{2} = 5.97.$$

The lattice value ([Bali 2001](#), pure-gauge $\text{SU}(3)$): $\sigma/\Lambda^2 = 6.25 \pm 0.5$. The match is 5% with zero free parameters. The one-loop correction (Supplement S, §5.3)—from the Seeley–DeWitt $a_2 = 4/135$ coefficient, the zero-mode Jacobian on $\mathbf{H}^2$, and the monopole–monopole interaction—shifts the tree-level value to $\sigma/\Lambda^2 = 6.46$, within the lattice error bar.

7.3 $\text{SU}(3)$ from the McKay correspondence at $N = 7$

Derivation 7.5 ($\text{SU}(3)$ gauge symmetry from the Klein quartic). The $N = 7$ vortex system determines the $\text{SU}(3)$ gauge symmetry — representation structure and dynamics — through the Klein quartic $X(7) = \Gamma(7) \backslash \mathbf{H}^2$:

- (a) Representation structure. $\text{PSL}(2, \mathbb{F}_7)$ embeds faithfully in $\text{SU}(3)$ via $\chi_3 = H^0(X(7), \Omega^1)$, with $\chi_3 \otimes \bar{\chi}_3 = \chi_1 \oplus \chi_8$.
- (b) Gauge dynamics. $\text{SU}(3)$ at the minimal compact level $k = 1$ defines a Chern–Simons gauge theory on the Seifert manifold. By the CS/WZW correspondence [[Witten, 1988](#)], the CS theory on a manifold with boundary induces the $\text{SU}(3)_1$ WZW model on the boundary, whose Kac–Moody currents $J^a(z)$ ($a = 1, \dots, 8$) are the dynamical gluons (conformal weight $(1, 0) = \text{spin } 1$).

Part (a) is representation theory; part (b) is the Witten CS/WZW theorem [[Witten, 1989](#)].

Proof of part (a): representation structure. Five algebraic steps.

1. *Frobenius orbits.* The multiplicative group $(\mathbb{Z}/7\mathbb{Z})^* \cong \mathbb{Z}/6\mathbb{Z}$ has a unique subgroup of order 3: $\langle 2 \rangle = \{1, 2, 4\}$ (equivalently $\langle 4 \rangle$, since $4 = 2^2$; the generator is not a choice — the subgroup is unique). The Frobenius action $\sigma: m \mapsto 2m \bmod 7$ has order 3, giving orbits $O_+ = \{1, 2, 4\}$, $O_- = \{3, 5, 6\}$. The other subgroups of $(\mathbb{Z}/7\mathbb{Z})^*$ do not give the needed structure: $\mathbb{Z}/2\mathbb{Z} = \{1, 6\}$ acts by $m \mapsto -m$ (the palindromic involution, already identified with conformal inversion in Paper I); $\mathbb{Z}/6\mathbb{Z}$ is the full group, giving a single orbit of size 6 with no complementary decomposition into $\mathbf{3} \oplus \bar{\mathbf{3}}$.

2. *Cyclic action.* On O_+ : $\phi_1 \mapsto \phi_2 \mapsto \phi_4 \mapsto \phi_1$.

3. *McKay embedding.* Diagonalise: $\eta_k = \frac{1}{\sqrt{3}} \sum_j \omega^{jk} \phi_{2j}$ ($\omega = e^{2\pi i/3}$). Then $\sigma(\eta_1) = \omega \eta_1$, $\sigma(\eta_2) = \bar{\omega} \eta_2$, so $\sigma = \text{diag}(\omega, \omega^{-1}) \in \text{SU}(2)$: the McKay embedding $\mathbb{Z}/3 \hookrightarrow \text{SU}(2) \rightarrow A_2 = \text{SU}(3)$ [[McKay, 1980](#), [González-Sprinberg and Verdier, 1983](#)].

4. *Klein quartic.* $X(7) = \Gamma(7) \backslash \mathbf{H}^2$ (genus 3, $\text{Aut} = \text{PSL}(2, \mathbb{F}_7)$, order 168). The differentials $H^0(X(7), \Omega^1)$ carry χ_3 ; σ generates $\mathbb{Z}/3 \subset \text{PSL}(2, \mathbb{F}_7)$.

5. *Eightfold Way.* $\chi_3 \otimes \bar{\chi}_3 = \chi_1 \oplus \chi_8$, $\text{Sym}^2(\chi_3) = \chi_6$, $\Lambda^2(\chi_3) = \bar{\chi}_3$. The adjoint χ_8 is irreducible under $\text{PSL}(2, \mathbb{F}_7)$. $\square$

Proof of part (b): gauge dynamics. Roadmap. Step B1 converts vortex displacements into CS connection perturbations (the dreibein chain). Step B2 extends this to all orders via the CS/WZW theorem. Steps B3–B4 apply the McKay correspondence and DHVW orbifold construction to *derive* the gauge group as $\text{SU}(3)$ at level 1 (see the derivation chain in §7.6 below). Step B5 promotes the boundary currents to 4D gauge bosons via KK reduction.

Step B1 (CS/WZW embedding). The charged Havelock modes η_1, η_2 (Step 3) are massive scalars on $\mathbf{H}^2$ with $m_k^2 = c_k(c_k - 1)$. Their boundary values parametrise $\mathbb{C}^2$, on which $\mathbb{Z}/3\mathbb{Z}$ acts as $\text{diag}(\omega, \omega^{-1}) \in \text{SU}(2)$. The CS/WZW theorem [Witten, 1988] gives the nonlinear identification: the boundary values of the CS gauge field A parametrise the WZW target manifold G , with the WZW action $S_{\text{WZW}}[g]$ equal to the on-shell CS action $S_{\text{CS}}[A|_{\partial}]$ for all field configurations, not just at the linearised level. The dreibein chain below verifies the linearised limit. A polygon vertex displacement δz_j perturbs the metric at the j -th cone point: $\delta g_{ab} = (\varepsilon_j/\pi) \partial_a \partial_b G(x, z_j) \delta z_j$. In the first-order formalism $g = e \otimes e$, this gives $\delta e^a = (1/2) e^b \delta g_{ab}$, and the CS perturbation is $\delta A^\pm = \delta \omega \pm \delta e/\ell$. The $\mathbb{Z}_N$ Fourier projection $\delta A_m = (1/N) \sum_j e^{-2\pi i j m/N} \delta A(z_j)$ decomposes into the mode- m CS perturbation, whose quadratic action is $|\delta A_m|^2 \lambda_m$ —the Havelock eigenvalue. The charged modes η_1, η_2 (Step 3) are the $m \in O_+$ Fourier projections of δA . The $\mathbb{Z}/3\mathbb{Z}$ invariance is exact at all orders because it is a property of the Casimir spectrum, not of the coupling strength: the Frobenius $\sigma: m \mapsto 2m \bmod 7$ permutes modes within orbits $O_+ = \{1, 2, 4\}$ and $O_- = \{3, 5, 6\}$, whose Casimir multisets are both $\{3, 5, 6\}$.

Step B3 (McKay + DHVW $\rightarrow \text{SU}(3)_1$ gauge sector). The McKay correspondence and the DHVW orbifold construction determine the gauge group and the CS level:

- *Gauge group.* The McKay correspondence for $\mathbb{Z}/3\mathbb{Z} \subset \text{SU}(2)$ gives the A_2 Dynkin diagram of $\text{SU}(3)$ [McKay, 1980, González-Sprinberg and Verdier, 1983]. The representation ring of $\mathbb{Z}/3\mathbb{Z} \subset \text{SU}(2)$ is isomorphic to the fusion ring of $\text{SU}(3)_1$ [Cappelli et al., 1987].
- *Level.* The DHVW construction [Dixon et al., 1985] applies to orbifolds of 2D CFTs by finite groups. The parent theory is the WZW boundary CFT at $c = 12b(N)$ (well-defined by the CS/WZW theorem [Witten, 1988]); the orbifold group is $\mathbb{Z}/3\mathbb{Z}$ (a symmetry of the WZW model, Step B4). *Consistency of the orbifold at irrational c .* The $\mathbb{Z}/3$ orbifold is a consistent (non-rational) CFT at $c = 12b(N)$. Each of the four consistency conditions is proved by a theorem that does not use rationality of c : (i) *Unitarity* (Friedan et al. 1984). The untwisted sector $\mathcal{H}^{\mathbb{Z}_3} \subset \mathcal{H}$ inherits positivity from the parent. For the twisted sectors: the Friedan–Qiu–Shenker theorem states that for $c > 1$, the Virasoro algebra has unitary highest-weight representations for *all* $h \geq 0$. The proof: the Kac determinant $\det M_n = \prod_{1 \leq r,s \leq n} (h - h_{r,s}(c))^{p(n-rs)}$ has zeros only at the Kac-table values $h_{r,s}(c)$, which are *complex* for $c > 1$ (the parametrisation $c = 1 - 6/(m(m+1))$ requires complex m). No real $h > 0$ intersects a Kac-table zero; therefore $\det M_n > 0$ for all n . At $(c, h) = (51.57, 2/9)$: level 1 Kac determinant $= 2h = 4/9 > 0$; level 2 Gram matrix has eigenvalues 26.7 and 1.21 (both positive; see Code Availability). (ii) *Modular invariance* (Dixon et al. 1985, Vafa 1986). The partition

function $Z_{\text{orb}} = (1/|G|) \sum_{g,h} Z_g^h$ transforms by the geometric rules $Z_g^h(\tau+1) = Z_g^{gh}$, $Z_g^h(-1/\tau) = Z_h^{g^{-1}}$. These follow from the torus path-integral with twisted boundary conditions and hold at every c . (iii) *OPE closure*. The $\mathbb{Z}/3$ action is an *inner* automorphism ($\sigma \in \text{PSL}(2, \mathbb{F}_7) \subset \text{Aut}(\text{CFT})$). Twisted-sector operators are spectral-flow images of untwisted operators. Spectral flow acts as an automorphism of the Virasoro algebra at every c (it shifts L_0 by a c -dependent constant; the commutation relations are c -independent). Since the OPE is generated by Virasoro descendants, $\sigma(O_1) \times \sigma(O_2) = \sigma(O_1 \times O_2) \in \text{OPE}$, and OPE closure holds at every c (including irrational c ; no finiteness of the primary spectrum is required). (iv) *No tachyons*. $h_k = k(3-k)/9 = 2/9 > 0$ (all $k = 1, 2$); the untwisted sector has $h \geq 0$. (v) *Partition function positivity*. The twisted-sector character is $\chi(c, h; q) = q^{h-c/24}/\eta(\tau)$, whose Fourier coefficients are partition numbers $p(n) \geq 0$. The untwisted-sector degeneracies $d_n = \dim(\mathcal{H}_n^{\mathbb{Z}_3}) \geq 0$ (dimension of an invariant subspace). Rationality of c would give a modular tensor category (finitely many primaries, braiding); at irrational c this structure is absent, but the CFT is well-defined: unitary, modular-invariant, with non-negative partition function—analogous to the free boson at irrational compactification radius. (vi) *Non-integrable sector is perturbative*. The four-point function of the Kac–Moody currents $\langle J^a J^b J^c J^d \rangle$, decomposed into Virasoro conformal blocks at $c = 12b(7) = 51.57$, receives 98.8% of its value from the integrable sector ($h = 0, 1, 2$) and 1.2% from non-integrable primaries (primarily the twist field at $h = 2/9$). The Kac–Moody OPE is universal (it depends on k and f^{abc} , not on c), so the singular structure of the gauge propagator is identical to rational c . The β -function coefficient $b_0 = 7$ is determined by the spectral decomposition on the boundary torus T^2 and the current Ward identity (which protects the anomalous dimension at any c), not by the value of c . The correction to the gauge dynamics from irrational c is bounded at 1.2%. The Cardy density of states $\rho(\Delta) \sim \exp(2\pi\sqrt{c\Delta/6})$ (Cardy 1986) applies to any modular-invariant partition function with a unique vacuum; the orbifold vacuum is unique (the identity is $\mathbb{Z}/3$ -invariant) and $h_k = 2/9 > 0$ introduces no new ground states. The Cardy constraint is therefore satisfied at every c , with $c_{\text{eff}} = c$. The DHVW construction produces $\hat{\mathbf{g}}_1$ for any $\Gamma \subset \text{SU}(2)$: the twist-field conformal dimension is $h_k = k(N_\Gamma - k)/N_\Gamma^2$, which depends only on the monodromy eigenvalue $e^{2\pi i k/N_\Gamma}$: the mode expansion around the twist point has fractional indices $n + k/N_\Gamma$, and the zeta-regularised zero-point energy gives $h = k/N_\Gamma - k^2/N_\Gamma^2$ exactly (a one-loop computation with no higher-order corrections). On the curved $\mathbf{H}^2$ background, the Liouville dressing shifts h_k by $O(h_k^2/c) = O(1/(N_\Gamma^4 c))$. At $N_\Gamma = 3$ and $c = 12b(7) \approx 52$: $\delta h = (2/9)^2/52 \approx 10^{-3}$, a 0.4% correction that does not affect the level identification. This fixes the current algebra level to 1.

The $\text{SU}(3)_1$ Chern–Simons theory is a *separate sector* from the gravitational $\text{SL}(2, \mathbb{R})$ CS theory, with its own central charge $c_{\text{SU}(3)} = 2$ (eq. (7.5) below). It is not the orbifold of the gravitational boundary theory (which has $c_{\text{grav}} = 12b(N)$ and remains unchanged): the orbifold construction determines *which* gauge group appears, and the DHVW construction determines the level, but the gauge sector is an independent CS theory on the same manifold. The compact gauge level $k = 1$ is independent of $k_{\text{grav}} = c/6 = 2b(N)$ (the non-compact gravitational coupling).

Step B4 (G_L invariance). The Frobenius $\mathbb{Z}/3\mathbb{Z}$ acts as an exact automorphism of the *full nonlinear* boundary CFT. The proof uses the $G_L \times G_R$ invariance of the WZW

action: left multiplication $g(x) \mapsto h \cdot g(x)$ by $h \in \mathbb{Z}/3\mathbb{Z} \subset G$ preserves both the kinetic term $\text{Tr}(g^{-1}\partial g \cdot g^{-1}\bar{\partial} g)$ and the Wess–Zumino term $\int \text{Tr}(g^{-1}dg)^3$ (both are manifestly G_L -invariant). This gives exact invariance of the action, the path integral, all n -point functions (Schwinger–Dyson), and all OPE coefficients. The Casimir multiset invariance follows: the Havelock eigenvalues are spectral data of the G_L -invariant theory. *Step B5 (CS/WZW correspondence $\rightarrow$ 4D gauge field)*. The chain from orbifold current algebra to 4D gauge theory has three links, each a theorem:

1. $\widehat{\mathfrak{su}}(3)_1$ current algebra on the boundary $\rightarrow \text{SU}(3)_1$ Chern–Simons theory in the bulk (Witten CS/WZW theorem [Witten, 1988]);
2. CS_3 on $\mathbf{H}^2 \times S^1 \rightarrow \text{YM}_2$ on $\mathbf{H}^2$ (KK reduction: the CS action $\int \text{Tr}(A \wedge dA + \frac{2}{3}A^3)$ on $\mathbf{H}^2 \times S^1$, after Fourier expanding $A = A^{(0)} + \sum_{n \neq 0} A^{(n)} e^{in\theta}$ and integrating over S^1 , yields the 2D YM action $\int_{\mathbf{H}^2} \text{Tr}(F^2)$ for the zero mode $A^{(0)}$, with coupling $g^2 = 4\pi/k$ and massive KK modes $M_n = n/R_{S^1}$);
3. YM_2 on $\mathbf{H}^2 + \text{WDW time direction} \rightarrow \text{YM}_{3+1}$: the breathing mode $\rho(t)$ (the scale factor of the $\mathbf{H}^2$ slice) satisfies the WDW equation $H\Psi = 0$ (Paper III, Proposition 3.7). The Feynman–Kac theorem identifies the WDW evolution parameter with Euclidean time (Paper III, §13): the WKB solution $\Psi_0 \propto e^{-\int V d\rho}$ evolves monotonically in ρ , so ρ is a good time coordinate. The gauge field $A^{(0)}$ on the spatial $\mathbf{H}^2$ thereby acquires the time direction ρ , giving YM_{2+1} . The KK tower $A^{(n)}$ with $n \geq 1$ provides the additional spatial dimension (the S^1 fiber), completing $1 + 2 + 1 = 4$ dimensions.

The $\text{SU}(3)$ gauge group is therefore *determined* by the Frobenius action on the Havelock modes (via McKay + DHVW) from the same inputs (H1)–(H3). In the topological bulk, there are zero propagating bulk modes; the boundary currents $J^a(z)$ ($a = 1, \dots, 8$, conformal weight $(1, 0)$, spin 1) are the dynamical degrees of freedom. After KK reduction on S^1 , the boundary currents become 4D spin-1 gauge bosons with the $\text{SU}(3)$ vertex structure inherited from the Kac–Moody OPE, at central charge

$$c_{\text{SU}(3)} = \frac{k \dim(\text{SU}(3))}{k+h^\vee} = \frac{8}{4} = 2. \quad (7.5)$$

The total boundary central charge is $c_{\text{tot}} = 12b(N)$. The gauge contributions ($c_{\text{SU}(3)} = 2$, $c_{\text{SU}(2)} = 1$, $c_{\text{U}(1)} = 1$) are subsumed: $c_{\text{grav}} = 12b(N) - 4$, with $G = 3/(2c_{\text{tot}}) = 3/(2 \times 12b(N))$ using the total central charge in the Brown–Henneaux relation. $\square$

Independent confirmation (ALE resolution). The K -matrix approach [Wen, 1995, Kronheimer, 1989] provides a cross-check: the intersection matrix of the resolved $\mathbb{C}^2/\mathbb{Z}_3$ is the A_2 Cartan matrix, which by the Wen equivalence also yields $\text{SU}(3)_1$.

7.4 Strong CP problem: $\theta_{\text{QCD}} = 0$ from level quantisation

The QCD vacuum angle is $\theta_{\text{QCD}} = 0$ exactly, by the following three-step argument.

1. *CS level quantisation.* For a compact gauge group, $\pi_3(\text{SU}(3)) = \mathbb{Z}$ forces the CS level $k \in \mathbb{Z}^+$. At $k = 1$: the vacuum angle is $\theta = 2\pi k_{\text{frac}} = 0$ (integer level, no fractional part).

2. *Fermion shifts preserve integrality.* Each massive Dirac fermion in representation R shifts the CS level by $\Delta k = T(R) \text{sign}(m)$ (parity anomaly; Redlich 1984), where $T(R)$ is the Dynkin index ($T(\text{fund}) = 1/2$ for $\text{SU}(3)$). For 6 quark flavors in the fundamental: $\Delta k = 6 \times T(\text{fund}) \times \text{sign}(m) = 6 \times 1/2 = 3 \in \mathbb{Z}$ (the three colours enter through T , not as independent fermions). All masses have the same sign (from a single Higgs VEV), giving $k_{\text{eff}} = 1 + 3 = 4$ (integer).
3. *Gravitational sector decouples.* The gravitational CS level $k_{\text{grav}} = 2b(N) = 8.60$ is irrational, with fractional part $k_{\text{frac}}^{\text{grav}} = 0.60$. This does not affect θ_{QCD} because the gauge and gravitational traces are orthogonal: $\text{Tr}(T_a^{\text{SU}(3)} T_b^{\text{SL}(2, \mathbb{R})}) = 0$.

At all loop orders, $k_{\text{eff}}^{\text{SU}(3)} \in \mathbb{Z}$ and $\theta_{\text{QCD}} = 0$. The mechanism is topological (level quantisation from π_3), requires no axion, and predicts: vanishing neutron EDM at all orders, null results at ADMX and ABRACADABRA (no axion field, no axion dark matter).

Orbit	Modes	Casimirs f	σ -eigenvalue
O_+	$\{1, 2, 4\}$	$\{3, 5, 6\}$	$1, \omega, \omega^2$
O_-	$\{3, 5, 6\}$	$\{6, 5, 3\}$	$1, \omega, \omega^2$

The QR/QNR orbits $O_{\pm}$ are the $\chi_3/\bar{\chi}_3$ sectors of Theorem 7.5: modes $\{1, 2, 4\}$ (quadratic residues mod 7) carry χ_3 , modes $\{3, 5, 6\}$ (non-residues) carry $\bar{\chi}_3$.

Consistency check ($C_F = 1 + 2B_2$). The fundamental Casimir $C_F = (N_c^2 - 1)/(2N_c) = 4/3$ at $N_c = 3$ equals $1 + 2B_2$, where $2B_2 = 1/3$ is the Laurent constant of the csc^2 kernel (Paper I, §7.6). $C_F = 1 + 2B_2$ holds *only* at $N_c = 3$ (the unique positive root of $3N_c^2 - 8N_c - 3 = 0$), which is the value determined by the Frobenius order $\text{ord}_7(2) = 3$. The identity $2B_2 = 1/N_c$ connects the csc^2 self-interaction constant to the gauge group rank: the *same kernel* that determines the self-interaction (B_2) also determines the gauge group (N_c) via the Frobenius orbits. This is a consistency check between two independently derived quantities, not a derivation of either.

7.5 U(1) from Kaluza–Klein

The S^1 fiber with flux $N/2$ gives a $\text{U}(1)$ gauge field at all N , with coupling $\alpha = 6/(\pi c)$ and the exact lock $\alpha/(8\pi G) = 1/(2\pi^2)$. This is standard Kaluza–Klein reduction. The KK $\text{U}(1)$ is the *only* abelian gauge symmetry in the theory: every KK mode carries charge $Q = m/N$ fixed by its S^1 momentum quantum number. This $\text{U}(1)$ *is* hypercharge: it is the only abelian gauge symmetry in the theory, and the charge normalisation $Q = m^*/N = 1/2$ at the critical $N = 4$ mode is the Standard Model hypercharge of the Higgs doublet—derived from the KK spectrum, not identified after the fact.

7.6 Summary: the derivation chain

The gauge group $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ at levels $k = 1$ is *derived* from the Seifert geometry $\mathbf{H}^2 \times_N S^1$, not identified after the fact. The chain uses three mathematical theorems applied to the geometry:

1. $\text{SU}(2)$ *from CS gravity* (Witten 1988): KK reduction on S^1 gives 2+1D gravity = $\text{CS}[\text{SU}(2)]$. At $N = 4$, the critical Casimir $f(2, 4) = 2 = j(j+1)$ with $j = 1$ (the adjoint). The Euler class $e = N/2$ breaks $\text{SU}(2)_R$ via the parity anomaly.

2. *SU(3) from McKay* (McKay 1980): The Frobenius at the smallest rational prime $p = 2$ (the canonical generator of the decomposition group at $(2) \subset \mathbb{Z}[\zeta_7]$) acts as $\sigma: a \rightarrow 2a \pmod{7}$ and has $\text{ord}_7(2) = 3$. The McKay correspondence maps $\mathbb{Z}/3 \subset \text{SU}(2) \rightarrow A_2 \rightarrow \text{SU}(3)$. The DHVW twist-field OPE (1985) fixes the level $k = 1$ (up to corrections $O(h^2/c) \sim 0.1\%$, which are small enough that integer quantisation of the CS level forces $k = 1$ exactly).
3. *U(1) from KK*: The fiber gives a U(1) gauge field with $K = 1$ (single-valued holonomy). Charge $Q = m/N$ is fixed by the S^1 momentum quantum number.

The CS-Havelock identity (Paper I, Theorem A.2) proves that the Casimir $f(m, N) = m(N-m)/2$ is simultaneously the vortex eigenvalue, the $\mathfrak{sl}(2, \mathbb{R})$ Casimir (Helgason), and the CS Wilson line Casimir (Etingof–Frenkel–Kirillov). These are the *same operator* (the Laplacian on $\mathbf{H}^2$) restricted to the same subspace ($\mathbb{Z}_N$ tangential modes).

Central charge budget (no-hidden-sector theorem): An exhaustive enumeration of simple Lie groups G at level $k \leq 3$ yields 33 combinations with total WZW central charge $c_{\text{gauge}} = 4$.¹ Of these, $\text{SU}(3)_1 \times \text{SU}(2)_1 \times \text{U}(1)_1$ is the unique one containing all three derived factors. Since $c_{\text{gauge}} = 2 + 1 + 1 = 4$ is fully accounted for, no additional gauge sector can fit within the central charge budget: the gauge group is *exactly* $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$, with no hidden factors.

Remark 7.2 (Structural inputs beyond N and M_P). *The derivation chain uses three mathematical theorems (Witten, McKay, DHVW) and one physical principle (Onsager selection, (H2b) from Paper III). The Frobenius base $p = 2$ is canonical (smallest prime coprime to N), not a free choice. The assignment of KK pairs to fermion generations and the identification of pair 3 with the Higgs doublet follow from the BF-crossing structure (only pair 3 has $c_{\text{up}} < 1 < c_{\text{dn}}$). The CKM phase $\delta = 70.2^\circ$ is a mathematical computation on $\mathbf{H}^2$; its physical interpretation as the CP-violating phase relies on the vortex-gravity isomorphism of Paper III.*

8 The Weinberg angle

Derivation 8.1 (Weinberg angle from the KK-reduced gauge action). *The KK reduction of CS_3 on $\mathbf{H}^2 \times S^1$ gives YM_2 on $\mathbf{H}^2$ with coupling $g^2 = 2\pi/((k+h^\vee)R)$ (Blau–Thompson 1993). The effective coupling $\alpha_{\text{eff}}(R) = g^2 C_2(R) = 2\pi h_R/R$ (with h_R the WZW conformal weight), and the common factor $2\pi/R$ cancels in the mixing angle ratio. The electroweak mixing angle is therefore:*

$$\sin^2 \theta_W = \frac{h_Y}{h_Y + h_W} = \frac{Q^2(k+h^\vee)}{Q^2(k+h^\vee) + K j(j+1)} = \frac{3}{11} \approx 0.2727, \quad (8.1)$$

where $h_Y = Q^2/k_Y$ and $h_W = j(j+1)/(k+h^\vee)$, with $k_Y = 1$ the $U(1)$ compact-boson level (distinguished from K for Gaussian curvature).

Proof. Three steps: derive the quantum numbers from the Casimir, derive the CS gauge couplings from the DHVW levels, compute the mixing angle from the KK-reduced action.

Step 1 (Quantum numbers from the Havelock Casimir). The critical mode at $N = 4$ has Havelock Casimir $f(2, 4) = m^*(N-m^*)/2 = 2$. This determines two quantum numbers:

¹The 33 combinations include $\text{SU}(5)_1$, $\text{SO}(8)_1$, $\text{SU}(3)_3$, and 30 multi-factor products; the full list is in the companion code (`gauge_group_derivation.py`).

- SU(2) Casimir: $C_2(j) = j(j+1) = f(m^*, N) = 2$, giving $j = 1$.
- U(1) charge: $Q = m^*/N = 2/4 = 1/2$.

Both are read off from the vortex spectrum. No choice is involved. The identification $C_2^{\text{SU}(2)} = f(m^*, N)$ is the Helgason symmetric space theorem (Proposition III-2.1): on $\mathbf{H}^2 = \text{SL}(2, \mathbb{R})/\text{SO}(2)$, the $\mathfrak{sl}(2, \mathbb{R})$ Casimir equals the Laplacian, which restricted to the m -th $\mathbb{Z}_N$ mode gives $m(N-m)/2$.

Step 2 (CS gauge couplings). At the Chern–Simons threshold [Witten, 1988]:

- SU(2): $g_W^2 = 4\pi/(k+h^\vee) = 4\pi/3$ ($k = 1$, $h^\vee(\text{SU}(2)) = 2$);
- U(1): $g_Y^2 = 4\pi/K = 4\pi$ ($K = 1$).

The level $k = K = 1$ is forced by the orbifold construction at *both* gauge groups—not by minimality:

- SU(3): the Frobenius at $N = 7$ generates $\mathbb{Z}/3\mathbb{Z}$; the DHVW twist-field OPE produces $\widehat{\mathfrak{su}}(3)_1$ at level 1 (Theorem 7.5, Step B3).
- SU(2): the Frobenius at $N = 4$ generates $\mathbb{Z}/2\mathbb{Z}$ (since $3^2 \equiv 1 \pmod{4}$, $\sigma: m \mapsto 3m$ has order 2). The McKay correspondence for $\mathbb{Z}/2\mathbb{Z} \subset \text{SU}(2)$ gives the A_1 Dynkin diagram = SU(2). The DHVW construction produces $\widehat{\mathfrak{su}}(2)_1$ at level 1.
- U(1): $K = 1$ is fixed by single-valued holonomy on S^1 . In the S^3 regime, $K = e = 1$ from the Euler class of the Hopf bundle; in the Seifert regime, $K = e/N$ from the orbifold normalisation.

In all three cases, $k = 1$ follows from the DHVW orbifold construction, independently of the parent theory’s level. The mechanism: adjacent twist fields $\sigma_k \times \sigma_{k+1}$ produce Kac–Moody currents $J^a(z)$ at conformal weight $h = 1$. The current–current OPE $J^a(z) J^b(w) \sim k \delta^{ab}/(z-w)^2 + \dots$ has level k determined by the twist-field normalisation, which for the standard DHVW construction gives $k = 1$ for any $\Gamma \subset \text{SU}(2)$ [Dixon et al., 1985]. The parent central charge $c = 12b(N)$ enters the gravitational sector but does not affect the gauge level. As a cross-check: $k = 2$ would give $\sin^2 \theta_W = 1/3$, and $k = 3$ gives $5/13$; both are experimentally excluded.

Step 3 (Mixing angle from KK-reduced Yang–Mills couplings). The KK reduction of CS_3 on $\mathbf{H}^2 \times S^1(R)$ at level k gives YM_2 on $\mathbf{H}^2$ with bare coupling $g^2 = 2\pi/((k+h^\vee)R)$ (Blau and Thompson 1993; the shift $k \rightarrow k+h^\vee$ is the standard one-loop CS renormalisation). The gauge–matter vertex couples with strength $g \times T_R^a$ (the Lie-algebra generator in representation R), so the *effective* coupling for matter in representation R is

$$\alpha_{\text{eff}}(R) = g^2 \times C_2(R) = \frac{2\pi C_2(R)}{(k+h^\vee)R}. \quad (8.2)$$

The Lie-algebra Casimir $C_2(R) = \sum_a (T_R^a)^2$ is a property of the representation, independent of k . The fibre radius R cancels in the mixing angle ratio:

$$\sin^2 \theta_W = \frac{g_Y^2 C_2(Y)}{g_Y^2 C_2(Y) + g_W^2 C_2(W)} = \frac{Q^2/k_Y}{Q^2/k_Y + j(j+1)/(k+h^\vee)}. \quad (8.3)$$

The numerator and denominator are products of two independent factors: the YM coupling $g^2 = 2\pi/((k+h^\vee)R)$ (determined by the CS level via KK reduction) and the Lie-algebra Casimir C_2 (a property of the representation, independent of k). Substituting from Steps 1–2 ($Q^2 = 1/4$, $K = 1$, $j = 1$, $k+h^\vee = 3$):

$$\sin^2 \theta_W = \frac{1/4}{1/4 + 2/3} = \frac{3}{11}.$$

□

Proposition 8.2 (Gauge coupling from the Lie-algebra Casimir at $k = 1$). *The $SU(2)$ gauge coupling in eq. (8.3) uses $C_2(j=1) = 2$, where $j = 1$ exceeds the integrable bound $j \leq k/2 = 1/2$ of $SU(2)_1$. The gauge coupling formula remains valid because the Lie-algebra Casimir and the TQFT spectrum are independent structures:*

1. The Casimir is level-independent. *The coupling to matter in representation R is $g \cdot T_R^a$ with $C_2(R) = \sum_a (T_R^a)^2 \in \mathfrak{su}(2)$. This is a property of the Lie algebra, not of the CS level k . The gauge coupling $g^2 = 4\pi/k$ (from KK reduction) times C_2 gives the interaction strength; both factors are k -independent. The mixing ratio $\sin^2 \theta_W = h_Y/(h_Y + h_W)$ uses $h_W = C_2(j=1)/(k+h^\vee) = 2/3$, which is a WZW conformal weight evaluated at the non-integrable representation $j = 1$. The conformal weight formula $h(j) = j(j+1)/(k+h^\vee)$ is the analytic continuation of the WZW conformal weight to arbitrary j ; it is an algebraic function of the Lie-algebra Casimir and the renormalized level, well-defined for any j (including $j > k/2$) because it does not require the representation to appear as a TQFT state. The level shift $k \rightarrow k+h^\vee = 3$ from the one-loop renormalization is included in the formula.*
2. The integrable bound restricts TQFT states, not gauge bosons. *The bound $j \leq k/2$ constrains anyonic quasiparticles (well-defined braiding and fusion). Gauge bosons propagate through the YM kinetic term (from KK reduction of the Seifert metric), not through topological braiding. The same situation holds in $SU(3)_1$: gluons transform in the adjoint ($C_2 = 3$, not integrable at $k = 1$), yet the $SU(3)$ gauge dynamics is standard.*
3. Zero quantum dimension confirms the separation. *At $k = 1$: $d_1 = \sin(3\pi/3)/\sin(\pi/3) = 0$. The $j = 1$ representation is topologically invisible (zero quantum dimension) but contributes to the YM dynamics through the Lie-algebra Casimir. The value $j = 1$ is fixed by the vortex Casimir: $f(m^*, 4) = 2 = j(j+1)$.*

Comparison with unification frameworks.

Framework	Definition at threshold	Value
SM alone	$g'^2/(g^2 + g'^2)$	free parameter
$SU(5)$ GUT	$\text{Tr}(Y^2)/\text{Tr}(T_3^2 + Y^2)$	$3/8 = 0.375$
CS/WZW (this work)	$h_Y/(h_Y + h_W)$	$3/11 = 0.273$

In 4D, the gauge coupling is representation-independent, so the mixing angle requires an external normalisation condition (the GUT embedding). In CS theory, the coupling h_R already incorporates the representation through (8.2), so the mixing angle is determined by the CS data alone—no GUT embedding is needed.

At the CS threshold there is a single matter species (the Havelock critical mode at $N = 4$). Below the polygon scale the theory flows to the 4D SM, and 3/11 becomes the boundary condition for the standard RG running (one-loop, $b_1 = 41/10$, $b_2 = -19/6$), which brings it to $\sin^2 \theta_W(M_Z) = 0.231$ —matching the measured value 0.2312 to 0.1%. This determines $M_{\text{poly}} \approx 300$ TeV: a consistency check (the running must land in the 50–500 TeV window). The SU(5) prediction 3/8 matches at $M_{\text{GUT}} \sim 10^{14}$ GeV (62% above experiment at tree level).

8.1 Spectral derivation of the electroweak couplings

The CS inverse couplings that enter the Weinberg angle can be derived independently from the spectral structure of the $\tilde{E}_8$ Cartan matrix (Paper I, §8; Paper V, §??). This provides a graph-theoretic derivation of the coupling constants that enter (8.3), without invoking Chern–Simons theory directly.

Lemma 8.3 (Spectral–CS bridge). *The spectral denominators $\{1, 2, 3, 5\}$ of the $\tilde{E}_8$ Cartan matrix are the stabiliser orders of the icosahedral symmetry axes (trivial, edge $\mathbb{Z}_2$, face $\mathbb{Z}_3$, vertex $\mathbb{Z}_5$). The dual Coxeter numbers of the SM gauge groups equal the corresponding stabiliser orders: $h^\vee(\text{SU}(3)) = 3 = |\mathbb{Z}_3|$ (face), $h^\vee(\text{SU}(2)) = 2 = |\mathbb{Z}_2|$ (edge), $h^\vee(\text{U}(1)) = 0$ (trivial). Consequently, the CS inverse couplings at level $k = 1$ are*

$$1/g^2(\text{SU}(3)) = 1 + 3 = 4, \quad 1/g^2(\text{SU}(2)) = 1 + 2 = 3, \quad k_Y = 1. \quad (8.4)$$

The vertex stabiliser $\mathbb{Z}_5$ gives $h^\vee(\text{SU}(5)) = 5$ (the GUT group), which breaks to $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ at the $K = 0$ transition.

Proof. The spectral denominators $k = h/\text{gcd}(m, h)$ for the affine m -values are: $k = 1$ (boundary, $m \in \{0, 30\}$), $k = 2$ (Majorana, $m = 15$), $k = 3$ (triangle, $m \in \{10, 20\}$), $k = 5$ (golden, $m \in \{6, 12, 18, 24\}$).

Each prime $p \mid h = 30$ corresponds to an icosahedral rotation axis with stabiliser $\mathbb{Z}_p$: $p = 5$ (vertices, 12 elements of order 5 or 10), $p = 3$ (faces, 20 elements of order 3 or 6), $p = 2$ (edges, 30 elements of order 4). The orbifold gauge group at each axis is $\text{SU}(p)$ via the McKay correspondence $\mathbb{Z}_p \subset \text{SU}(2) \rightarrow \tilde{A}_{p-1}$, with $h^\vee(\text{SU}(p)) = p$.

For the SM gauge groups after breaking: $\text{SU}(3)$ retains $h^\vee = 3$ from the face stabiliser, $\text{SU}(2)$ retains $h^\vee = 2$ from the edge stabiliser, and $\text{U}(1)$ is abelian ($h^\vee = 0$). The universal CS level $k = 1$ (Hopf Euler class, §8) gives (??). $\square$

Corollary 8.4 (Spectral Weinberg angle). *The spectral denominators of the electroweak sectors recover*

$$\sin^2 \theta_W = \frac{Q^2/k_Y}{Q^2/k_Y + j(j+1)/(k+h^\vee)} = \frac{1/4}{1/4 + 2/3} = \frac{3}{11} = \frac{a-1}{a+N_{\text{crit}}}, \quad (8.5)$$

where $a = k_1(E_8) = 4$ and $N_{\text{crit}} = 7$. The numerator $a - 1 = 3 = \dim(\text{SO}(3))$ is the spatial rotation dimension; the denominator $a + N_{\text{crit}} = 11$ is the sum of all spectral denominators $\{1, 2, 3, 5\}$.

Proof. By the spectral–CS bridge (Lemma 8.3), the electroweak couplings in (8.3) are $k + h^\vee(\text{SU}(2)) = 3$ and $k_Y = 1$. With $Q = 1/2$ and $j = 1$ from the Havelock Casimir (Step 1, §8), the result follows. $\square$

Proposition 8.5 (Bernoulli–Coxeter identity). $B_4 = B_8 = -1/30 = -1/h(E_8)$. Consequently $\zeta(-3) = 1/120 = 1/|I^*|$ and $\zeta(-7) = 1/240 = 1/|\text{roots}(E_8)|$.

Proof. By von Staudt–Clausen, $\text{denom}(B_{2k}) = \prod_{(p-1)|2k} p$. For B_4 : $(p-1)|4$ gives $p \in \{2, 3, 5\}$, $\text{denom} = 30$. For B_8 : $(p-1)|8$ gives $p \in \{2, 3, 5\}$ (since $p = 9 = 3^2$ is not prime), $\text{denom} = 30$. Both numerators are -1 by the Bernoulli recursion. Then $\zeta(-3) = -B_4/4 = (1/30)/4 = 1/120$ and $\zeta(-7) = -B_8/8 = (1/30)/8 = 1/240$. $\square$

Remark 8.1. The identity $B_4 = B_8$ relies on $\text{rank}(E_8) + 1 = 9 = 3^2$ being composite. If 9 were prime, $\text{denom}(B_8) = 270 \neq 30$, and the Riemann zeta would not encode $|I^*|$ at $s = -3$. The number-theoretic consistency of the framework requires $\text{rank} + 1 = p^2$ for some $p \mid h$.

9 The action

The complete Havelock field theory action on the spacetime $M_4 = \mathbb{R} \times (\mathbf{H}^2 \times_N S^1)$ is:

$$S = S_{\text{grav}} + S_{\text{CS}} + S_{\text{gauge}} + S_{\text{scalar}} + S_{\text{ferm}} + S_{\text{Yuk}}. \quad (9.1)$$

Gravitational sector. After KK reduction on S^1 , the 2+1D gravitational action is the Chern–Simons action:

$$S_{\text{CS}} = \frac{k}{4\pi} \int_{M_3} \text{Tr} \left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right), \quad k = \frac{c}{6} = 2b(N), \quad (9.2)$$

with A the $\text{SU}(2)$ (Euclidean) gauge field.

U(1) gauge sector.

$$S_{\text{gauge}} = -\frac{1}{4\alpha} \int d^4x \sqrt{-g} F_{\mu\nu} F^{\mu\nu}, \quad \alpha = \frac{6}{\pi c}. \quad (9.3)$$

Scalar sector (KK modes). N charged scalars Φ_m ($m = 0, \dots, N-1$) with KK masses $\mu_m = |m - N/2|$:

$$S_{\text{scalar}} = \int d^4x \sqrt{-g} \sum_{m=0}^{N-1} \left[|D_\mu \Phi_m|^2 - \mu_m^2 |\Phi_m|^2 \right], \quad (9.4)$$

where $D_\mu \Phi_m = (\partial_\mu - imA_\mu) \Phi_m$.

Fermion sector. N Dirac fermions Ψ_m with masses $\mu_m^f = |m - (N-1)/2|$:

$$S_{\text{ferm}} = \int d^4x \sqrt{-g} \sum_{m=0}^{N-1} \bar{\Psi}_m (i \not{D} - \mu_m^f) \Psi_m. \quad (9.5)$$

Yukawa sector.

$$S_{\text{Yuk}} = \int d^4x \sqrt{-g} \sum_{i,j} Y_{ij} \bar{\Psi}_i \Phi \Psi_j + \text{h.c.}, \quad (9.6)$$

with the selection rule $m_i - m_j + m_H \equiv 0 \pmod{N}$ (KK charge conservation on S^1).

All couplings are determined by $c = 12b(N)$.

The 4D effective Lagrangian. At energies $E \ll M_{\text{poly}}$, the KK modes (graviphoton, radion, and the KK tower) decouple, and the effective theory on $\mathbb{R}^{3,1}$ is:

$$\begin{aligned} \mathcal{L}_{4\text{D}} = & -\frac{1}{16\pi G_4}(R - 2\Lambda) && [\text{gravity, from boundary } T] \\ & -\frac{1}{4}\sum_a F_{\mu\nu}^a F^{a,\mu\nu} && [\text{SU}(3)\times\text{SU}(2)\times\text{U}(1), \text{ from boundary } J^a] \\ & +\bar{\Psi}(i\not{D} - YH)\Psi && [3 \text{ gen. fermions, } \mathbf{H}^2 \text{ radial profiles on } \mathbf{H}^2] \\ & +|D_\mu H|^2 - V(H) && [\text{Higgs, } v = 246 \text{ GeV, } \lambda_H = 0.122] \end{aligned} \quad (9.7)$$

All couplings in (9.7) are determined by $N = 7$ and M_P ; no free parameters remain.

Remark 9.1 (Unified 3+1D emergence from the boundary CFT). *The 3+1D theory is not assembled from separate mechanisms. It emerges from a single object: the boundary CFT at $c = 12b(N)$ on the torus $T^2 = \partial M$. The CS/WZW theorem encodes all bulk degrees of freedom in the boundary operators, organized by conformal weight:*

Boundary operator	Weight	4D field	Origin
Stress tensor $T(z)$	$(2, 0)$	Graviton $h_{\mu\nu}$	Virasoro
WZW currents $J^a(z)$	$(1, 0)$	Gauge bosons A_μ^a	Kac–Moody
KK primaries $\Phi_n(z)$	$(h_n, \bar{h}_n)$	Matter fields	Fiber modes
Modular parameter τ	—	Cosmological Λ	Partition fn

The graviton, gauge bosons, and matter are three sectors of this single boundary theory—not three independent constructions. The time direction emerges from the Wick rotation $t_E \rightarrow it_L$ of the Euclidean boundary CFT to Lorentzian signature, which is well-defined because the CFT is unitary (DHVW orbifold of a unitary WZW model). Lorentz invariance follows from the Einstein equations (Theorem 3.3).

10 UV completion

The theory is UV-complete in the following precise sense.

Non-perturbative definition. The boundary CFT is a $\mathbb{Z}_N$ orbifold theory at central charge $c = 12b(N)$, with known partition function, OPE coefficients, and conformal blocks. By the CS/WZW theorem [Witten, 1988], this boundary theory defines the bulk Chern–Simons theory at $k = c/6 = 2b(N)$ non-perturbatively. At leading order, $c = N(N+1) + O(\ln N)$; the subleading corrections $(-12\ln 2 + 12(\ln N)/(N-1))$ encode the regularised vortex self-energy. The c -theorem is satisfied: the free-field UV limit has $c_{\text{UV}} = N(N+1)$, and $c_{\text{IR}} = 12b(N) < N(N+1)$ for all $N \geq 3$. There is a single central charge throughout: $c = 12b(N)$ enters both the Brown–Henneaux relation ($G = 3/(2c)$) and the WDW breathing-mode equation.

Super-renormalisability. After KK reduction to 1+1D, the superficial degree of divergence is $D = 2 - E$ (E = external legs). Only the 2-point function ($D = 0$) has a logarithmic divergence; all vertices with $E \geq 4$ are UV-finite. Two renormalisation parameters suffice: the vortex core size a (UV cutoff for self-energy) and the central charge c (overall energy scale).

Physical observables. All physical observables—Havelock eigenvalues, partition function, couplings, cosmological constant, Weinberg angle—are exact (no perturbative approximation). The only divergences (self-energy at $d = 0$, KK zero-point sum) are renormalised by physical parameters within the framework.

The S^3 regime and E_8 (Paper V). The Seifert manifold $\mathbf{H}^2 \times_N S^1$ has Euler class $e = N/2$, giving the SM couplings at $K \leq 0$. For $K > 0$ (spatial S^3), the Hopf fibration $S^1 \rightarrow S^3 \rightarrow S^2$ has Euler class $e = 1$ (the first Chern class of $\mathcal{O}(1)$ on $\mathbb{CP}^1 = S^2$), fixing the CS level at $k = 1$. At this level, the full gauge group is E_8 with $1/g^2 = k + h^\vee = 31$ and WZW central charge $c = 248/31 = 8$ (exactly). The $E_8 \rightarrow$ SM breaking at $K = 0$ is a first-order topological phase transition: the Thurston geometry changes from S^3 to $\mathbb{R}^3$, the icosahedral ground state ($N = 12$, A_5 , E_8) dissolves, and the heptagonal ground state ($N = 7$, $\mathbb{Z}_7$, SM) takes over. No Higgs mechanism is involved. The 236 coset generators (E_8/SM , dimension $248 - 12$) acquire infinite mass at the transition. This is developed in full in Paper V.

KK spectrum and corrections. The KK tower on the Seifert manifold has mass-squared $M_{n,m}^2 = (\lambda_m + n^2) M_{\text{KK}}^2$ ($n \geq 0$, $1 \leq m \leq N-1$, $\lambda_m = f(m^*, N) - f(m, N)$). At $N = 11$ the lowest states are:

n	m	M/M_{KK}	Identity
0	5, 6	0	graviton (critical modes)
0	4, 7	1	lightest DM ($\lambda = 1$)
1	5, 6	1	first graviton KK
0	3, 8	$\sqrt{3}$	DM ($\lambda = 3$)
1	4, 7	$\sqrt{2}$	first DM ₁ KK
0	2, 9	$\sqrt{6}$	DM ($\lambda = 6$)
0	1, 10	$\sqrt{10}$	DM ($\lambda = 10$)

The lightest DM particle and the first graviton KK excitation are *mass-degenerate* at M_{KK} — a structural consequence of $\lambda_4 = 1 = 1^2$. The one-loop corrections from the full tower, summed via zeta regularisation ($\sum_{n=1}^{\infty} n^{-2} = \pi^2/6$, $\sum_{n=1}^{\infty} \ln n = -\zeta'(0) = \frac{1}{2} \ln(2\pi)$), are:

Observable	$M_{\text{KK}} = 50 \text{ TeV}$	$M_{\text{KK}} = 300 \text{ TeV}$
$\delta G/G$	2.5×10^{-7}	7×10^{-9}
$\delta g^2/g^2$	2.5×10^{-7}	7×10^{-9}
$\delta m_H^2/m_H^2$	1.5×10^{-7}	4×10^{-9}

All corrections are 10^{-7} – 10^{-9} , six to eight orders of magnitude below the geometric-tier accuracy and four to six orders below the algebraic tier. The boundary CFT controls the UV; the KK corrections are finite, calculable effects in the IR effective theory.

11 One-loop corrections

The Chern–Simons non-renormalisation theorem guarantees that the CS level receives *at most* a one-loop correction; higher loops are forbidden by large-gauge-transformation invariance.

Proposition 11.1 (One-loop structure).

1. **Coupling lock (all-orders exact).** The ratio $\alpha/(8\pi G) = 1/(2\pi^2)$ is algebraically c -independent: both G and α are proportional to $1/c$, so their ratio cancels c . The one-loop shift $c \rightarrow c - 3N$ (N Dirac fermions, each contributing $c = -3$ from the gravitational anomaly: -1 from the Dirac field -2 from the parity anomaly in odd dimensions) does not affect the ratio. This is exact at all orders.
2. **CS level shift.** Each massive 3D Dirac fermion shifts the CS level by $\text{sgn}(\mu_m)/2$ (parity anomaly). The KK spectrum $\mu_m = (m + \frac{1}{2} - N/2)/R$ is antisymmetric, so the massive modes cancel pairwise; for odd N , the single zero mode at $m = (N-1)/2$ contributes $\pm 1/2$ (sign from fiber orientation). The physical level is $k_{\text{phys}} = c/6 - 1/2$ (odd N) or $c/6$ (even N). All downstream quantities depend only on the fractional part $\kappa = \text{frac}(k_{\text{phys}}) = 0.096$ (at $N = 7$), which is unaffected by the integer shift $-(N-1)/2$. No further renormalisation occurs beyond one loop.
3. **Cosmological constant.** The 3D Coleman–Weinberg correction $\Delta\Lambda = -(1/12\pi)(\sum_b \mu_m^3 - \sum_f |\mu_m^f|^3)$ is a finite sum. At $N = 11$: $\Delta\Lambda/\Lambda_{\text{tree}} = -9.1\%$.
4. **Fermion mass ratios (quantum stable).** The one-loop self-energy from CS exchange gives a universal mass shift $\delta m/m = -1/(4\pi k)$ ($< 1\%$ for $N \geq 7$), identical for all KK modes. Mass ratios and Yukawa texture zeros are unchanged.
5. **Weinberg angle and the polygon scale.** The tree-level prediction $\sin^2 \theta_W = 3/11$ matches the experimental 0.2312 at M_Z via standard SM running for

$$M_{\text{poly}} \approx 300 \text{ TeV} \quad (10^{5.5} \text{ GeV}). \quad (11.1)$$

At $M_{\text{poly}} = 50 \text{ TeV}$ the prediction is $\sin^2 \theta_W(M_Z) = 0.240$ (3.9% above experiment); $O(1)$ threshold corrections at the polygon scale accommodate the 50–500 TeV range.

12 The fermion mass structure

This section derives the fermion spectrum from the Seifert geometry. Every result below follows from the $\mathbb{Z}_7$ orbifold structure, the $\mathbf{H}^2$ radial profiles, and the BF instanton on $\mathbf{H}^2$; no structural identifications are needed beyond those already established in §7.6. The results fall into three tiers: *Exact* (no adjustable parameters): the Yukawa texture zeros (§12.1–§12.3), the CKM phase $\delta = 70.2^\circ$ (§12.7), and sector decoupling (§12.5). *One-parameter*: the mass hierarchy (§12.4) and tree-level CKM (§12.6) depend on the warp factor $\sigma = \ln(M_{\text{poly}}/v)$, pinned by the Weinberg angle running; all mass *ratios* are then fixed. *Non-perturbative*: the orbifold image suppression (§12.8) gives $|V_{cb}| = 0.044$.

Derivation chain for the fermion sector. The complete chain from Seifert geometry to fermion masses:

1. $\mathbb{Z}_7$ charge conservation $\rightarrow$ Yukawa texture (4 zeros, 5 nonzero entries);
2. $N = 4$ isospin ($\mu_4 = 1 \mp T_3$) $\rightarrow$ up/down mass splitting;
3. KK spectrum $c_k = \sqrt{\mu_{7,k}^2 + \mu_4^2} \rightarrow$ conformal dimensions $\rightarrow$ RS profiles $f_k(\sigma)$;
4. Central charge $c = 12 b(7) \rightarrow$ instanton fugacity $\mathcal{K} = e^{-2\pi k_{\text{frac}}} = 0.548$;

5. BF instanton ($\Delta q = 2$) $\rightarrow Y_{33} = \mathcal{K}^2 f_3^2 \rightarrow m_s/m_b = 0.023$;
6. $\mathbf{H}^2$ Dirac scattering ($c : \frac{1}{2} \rightarrow \frac{3}{2}$) $\rightarrow \delta_{\text{CKM}} = \arg \Gamma(\frac{1}{2}+i) = 70.2^\circ$;
7. Orbifold image $Q_1(\cosh d)$ on $\mathbf{H}^2/\mathbb{Z}_7 \rightarrow |V_{cb}| = 0.044$.

Each step is a computation on the Seifert manifold. The only input beyond the gauge sector (§7.6) is $\sigma = \ln(M_{\text{poly}}/v)$, which is fixed by the Weinberg angle running.

12.1 The Yukawa texture from KK charge conservation [exact]

The Yukawa coupling Y_{ij} between fermion pairs i, j and the Higgs (pair 3, modes 3 and 4) is nonzero iff $m_i - m_j + m_H \equiv 0 \pmod{7}$. Evaluating for all pairs:

$$Y = \begin{pmatrix} 0 & Y_{12} & Y_{13} \\ Y_{21} & Y_{22} & 0 \\ Y_{31} & 0 & 0 \end{pmatrix}, \quad (12.1)$$

with four texture zeros at $(1, 1), (2, 3), (3, 2), (3, 3)$ —determined entirely by the arithmetic of $\mathbb{Z}/7\mathbb{Z}$.

12.2 The 2+1 mass pattern and the exact tree-level ratio [exact]

The matrix $YY^\dagger$ is block-diagonal: a 2×2 block (pairs 1,2) and a 1×1 block (pair 3). The eigenvalues of the 2×2 block (in the large- σ limit where the $\mathbf{H}^2$ radial profiles saturate) are $\lambda_\pm = (5 \pm \sqrt{13})/2$, giving the exact tree-level mass ratio

$$\left. \frac{m_1}{m_2} \right|_{\text{tree}} = \sqrt{\frac{5 + \sqrt{13}}{5 - \sqrt{13}}} \approx 2.48, \quad (12.2)$$

while the lightest mass $m_3 \propto e^{-\sigma/2}$ is exponentially suppressed by the warp factor σ .

12.3 Up–down splitting from the $N = 4$ isospin [exact]

The $\text{SU}(2)$ doublet structure from the $N = 4$ sector gives different KK masses for up-type ($\mu_4 = 1/2$) and down-type ($\mu_4 = 3/2$) fermions. The combined bulk mass $c_{k,I} = \sqrt{\mu_{7,k}^2 + \mu_{4,I}^2}$ separates the two types, with down-type having larger c (more UV-localised, hence lighter).

12.4 The fermion mass hierarchy [one-parameter: σ]

The tree-level texture (12.1) has $Y_{33} = 0$ (the charge conservation $3+3+3 = 9 \not\equiv 0 \pmod{7}$ forbids a direct b -quark Yukawa). At tree level, m_b comes from $Y_{31} = f_3 f_1$, which involves the most UV-localised profile f_1 . The Dirac zero-mode profile on the Seifert manifold is $\psi_c(\rho) = N_c (\sinh \rho)^{c-1/2}$ (from the KK-reduced Dirac equation with warp factor $A(\rho) = \ln(\sinh \rho)$). In the holographic regime $\rho \gg 1$, $\sinh \rho \approx e^\rho/2$ and the profile reduces to the Randall–Sundrum form $f_k = \sqrt{(2c_k-1)/(e^{(2c_k-1)\sigma}-1)}$ at warp factor $\sigma = \ln(M_{\text{poly}}/v)$, where σ is the geodesic distance on $\mathbf{H}^2$ between the holographic positions of v and M_{poly} . The profile suppresses f_1 exponentially ($c_1 = 2.50$), giving the tree-level ratio $m_s/m_b \approx 0.63$.

The *BF instanton resolves this*. The same BF-crossing instanton that generates $\delta_{\text{CKM}} = 70.2^\circ$ (Theorem 12.1) carries $\Delta q = 2$ units of $\mathbb{Z}_7$ charge and generates an effective Y_{33} :

$$Y_{33}^{\text{inst}} = \mathcal{K}^2 f_3^2, \quad \mathcal{K} = 0.548 \text{ (eq. (5.2))}. \quad (12.3)$$

Since $\mathcal{K}^2 f_3 \gg f_1$ at the physical warp factor ($\mathcal{K}^2 f_3/f_1 \approx 260$ at $\sigma = 7.1$), the instanton contribution dominates: $m_b \propto Y_{33}^{\text{inst}} = \mathcal{K}^2 f_3^2$. The s -quark mass comes from the tree-level $Y_{22} = f_2^2$ (charge-conserving, no instanton needed). The mass ratio is therefore

$$\frac{m_s}{m_b} = \frac{f_2^2}{\mathcal{K}^2 f_3^2} = \frac{1}{\mathcal{K}^2} \left(\frac{v}{M_{\text{poly}}} \right)^{2\Delta c}, \quad (12.4)$$

where $\Delta c = c_2 - c_3$ is the conformal-dimension gap between generations 2 and 3, with $c_k = \sqrt{\mu_{7,k}^2 + \mu_4^2}$ and $\mu_{7,k} = |k - (N-1)/2|$ (the pair-centred KK mass). At tree level $\Delta c_0 = \sqrt{1+9/4} - 3/2 = 0.303$. The Seifert fiber connection shifts the effective KK charge by $e/c = (N/2)/(12b(N))$ per unit of angular momentum (the same Euler class $e = N/2$ that drives parity violation in §7.1). Combined with the $O(1/c)$ BO anharmonic correction, the conformal-dimension gap becomes

$$\Delta c = \Delta c_0 + \frac{\mu_{7,2}}{c_2} \cdot \frac{1+e}{c} = 0.303 + 0.048 = 0.351. \quad (12.5)$$

The two corrections (Euler class and BO anharmonic) are additive because they act on different sectors: the Euler class shifts the KK charge (a geometric, fiber-connection effect), while the BO anharmonic shifts the radial potential (a spectral effect from the W_N quartic coupling). Their cross-term is $O(\mu_{7,2} e/c^3) = O(10^{-5})$, negligible at the precision of (12.5). Standard SM two-loop running (with $b_1 = 41/10$, $b_2 = -19/6$, and the two-loop matrix b_{ij} of Machacek and Vaughn 1984) gives $\sin^2 \theta_W = 3/11$ at $M_{\text{poly}} = 294$ TeV.² The resulting mass ratio: $m_s/m_b = 0.023$ (observed 0.024, 4% match). One calibration input (Δm_{atm}^2) pins M_{poly} within the predicted range; all other quantities are computed: $\mathcal{K}$ is fixed by the CKM phase, Δc by the KK spectrum, and M_{poly} by the Weinberg angle (with two-loop corrections).

The tree-level CKM has $|V_{cb}|_{\text{pert}} = 0.092$ ($2.2\times$ the observed 0.042). The non-perturbative CKM (§12.8) resolves this separately: J -conservation combined with the scattering phase $\delta = 70.2^\circ$ redistributes the mixing angles, giving $|V_{cb}|_{\text{np}} = 0.044$ (6% above observed).

The d -quark mass from CKM mixing (Gatto relation). The bare Yukawa eigenvalue $m_d^0 \propto f_1^2$ is doubly UV-suppressed and negligible ($m_d^0 \approx 0.002$ MeV at $\sigma = 7.1$). The physical d -quark mass comes from CKM mixing with the s quark:

$$m_d \approx |V_{us}|^2 m_s, \quad \frac{m_d}{m_s} = \sin^2 \theta_C. \quad (12.6)$$

With $|V_{us}| = 0.237$ (the tree-level Cabibbo angle): $m_d/m_s = 0.056$ (observed: 0.049, 14% off). This is the Gatto relation [Gatto et al., 1968]: no new parameter enters. The three down-type mass ratios are then: $m_s/m_b = 0.024$ (eq. (12.4)), $m_d/m_s = 0.056$ (eq. (12.6)), $m_d/m_b = 0.0013$ (product; observed 0.0011).

²The two-loop SM β -functions [Machacek and Vaughn, 1984] shift M_{poly} from 274 (one-loop) to 294 TeV (two-loop); see Code Availability.

12.4.1 The u -quark mass and the broad Higgs

The up-type conformal dimensions are $c_t = \sqrt{0+(1/2)^2} = 1/2$ (exactly BF-marginal: the top quark is conformally coupled), $c_c = \sqrt{1+1/4} = 1.12$, $c_u = \sqrt{4+1/4} = 2.06$. The BO ground state (Higgs profile) has width $\sigma_H = 1/\sqrt{V''(\rho^*)} = 2.74$, which exceeds $\rho^* = 1.73$: the Higgs wavefunction extends to $\rho \approx 0$, sampling the unsuppressed region where all RS profiles are comparable. The effective (1,2) mixing angles from the overlap integral $\theta_{12}^I = \int f_1^I |\Psi_H|^2 \sinh \rho d\rho / \int f_2^I |\Psi_H|^2 \sinh \rho d\rho$ are $\theta_{12}^{\text{down}} = 0.28$ and $\theta_{12}^{\text{up}} = 0.032$ (the up-sector mixing is $9\times$ smaller because $\Delta c_{\text{up}} = 0.94 > \Delta c_{\text{down}} = 0.70$). This gives simultaneously: $|V_{us}| \approx |\theta_d - \theta_u| = 0.25$ (observed 0.224), $m_d \approx \theta_d^2 m_s = 7.6 \text{ MeV}$ (observed 4.7), $m_u \approx \theta_u^2 m_c = 1.3 \text{ MeV}$ (observed 2.2). The 40–60% discrepancies on m_u , m_d reflect the sensitivity of light-quark masses to the Higgs profile tail and QCD running to 2 GeV; the qualitative prediction $m_u < m_d \ll m_s$ is robust and parameter-free.

12.5 Sector decoupling [exact]

The $N = 7$ and $N = 8$ polygon sectors are decoupled by the mod-56 arithmetic of the common overfield $\mathbb{Q}(\zeta_{56})$: the cross-sector Yukawa coupling requires $8(m_1 + m_2) \equiv 28 \pmod{56}$, which has no integer solution ($\gcd(8, 56) = 8$ does not divide 28). The texture zeros of the $N = 7$ Yukawa are topologically protected.

12.6 CKM mixing matrix [one-parameter: σ]

Roadmap. The CKM matrix has three layers of corrections: (a) *tree-level*: diagonalisation of the $\mathbf{H}^2$ Yukawa texture gives $|V_{cb}| = 0.092$, $\delta = 15^\circ$; (b) *non-perturbative phase*: the BF-crossing phase $\arg \Gamma(\frac{1}{2} + i)$ (Theorem 12.1) gives $\delta = 70.2^\circ$; (c) *orbifold image suppression*: the χ_8 channel of the $\text{PSL}(2, \mathbb{F}_7)$ decomposition gives the final $|V_{cb}| = 0.044$.

The CKM matrix $V = U_u^\dagger U_d$ is obtained by diagonalising the up-type ($\mu_4 = 1/2$) and down-type ($\mu_4 = 3/2$) mass matrices $M = v \times Y$ with the $\mathbf{H}^2$ radial profile overlaps. Predicted ($\mu_4 = 1 \mp T_3$ derived from the BF threshold and the Higgs T_3 eigenvalue; see Paper VI):

$$|V| \approx \begin{pmatrix} 0.972 & 0.237 & 0.006 \\ 0.236 & 0.967 & 0.092 \\ 0.016 & 0.091 & 0.996 \end{pmatrix} \quad \text{Observed: } \begin{pmatrix} 0.974 & 0.224 & 0.004 \\ 0.224 & 0.974 & 0.042 \\ 0.008 & 0.039 & 0.999 \end{pmatrix}$$

The Cabibbo angle $\theta_C = 13.7^\circ$ (observed 13.0° , 5% off). The hierarchy $|V_{us}| \gg |V_{cb}| \gg |V_{ub}|$ is predicted.

12.7 The CKM phase [exact]

Derivation 12.1 (CKM phase from the scattering phase on $\mathbf{H}^2$). *The CKM CP-violating phase is*

$$\delta_{\text{CKM}} = \frac{1}{2} \log \cosh(\pi) = 70.2^\circ. \quad (12.7)$$

The observed value is $\delta = 69^\circ \pm 3^\circ$ (PDG 2024); the discrepancy is 1.2° (40% of 1σ).

Proof. Four steps: determine which mode contributes, derive the spectral parameter from the KK structure, compute the overlap integral, evaluate the phase.

Step 1 (Only the BF-crossing mode contributes). For generation 3, mode $m = 3$ ($\mu_7 = 0$): the conformal dimension transitions from $c_{\text{up}} = \mu_4 - T_3 = \frac{1}{2}$ (up-type) to $c_{\text{dn}} = \mu_4 + T_3 = \frac{3}{2}$ (down-type), crossing the BF bound $c = 1$ ($M^2 = c(c-1) = 0$). All other modes have $c_{\text{up}} > 1$ (since $\mu_7 \geq 1$ gives $c \geq 1.12$); for these, both c_{up} and c_{dn} are real and positive, so $\arg \Gamma(c) = 0$ —no phase difference and no contribution to δ . The CKM phase comes entirely from generation 3.

Step 2 (The spectral parameter $m = 1$ from the conformal dimension sweep). The $\mathbf{H}^2$ radial profile at conformal dimension c is the conical (Mehler) function $P_{-1/2+is}(\cosh \rho)$ with spectral parameter

$$s = c - \frac{1}{2}. \quad (12.8)$$

This is the *standard parameterisation* of the Mehler function (DLMF §14.20): c and s are related by an additive constant.

As the conformal dimension sweeps from $c_{\text{up}} = \frac{1}{2}$ to $c_{\text{dn}} = \frac{3}{2}$ (the BF crossing of Step 1), the spectral parameter sweeps from $s_{\text{up}} = 0$ to $s_{\text{dn}} = 1$:

$$\Delta s = \Delta c = c_{\text{dn}} - c_{\text{up}} = \frac{3}{2} - \frac{1}{2} = 1.$$

The $\mathbf{H}^2$ scattering matrix for the Dirac operator at spectral parameter s is $S(s) = \Gamma(\frac{1}{2} + is)/\Gamma(\frac{1}{2} - is)$, with scattering phase $\arg S(s) = 2 \arg \Gamma(\frac{1}{2} + is)$. At $s = 0$: $S(0) = 1$ (zero phase; $\Gamma(1/2)$ is real). At $s = 1$: $S(1) = \Gamma(1/2 + i)/\Gamma(1/2 - i)$, which is well-defined (Γ has no poles on the line $\text{Re}(z) = 1/2$). The accumulated phase from $s = 0$ to $s = 1$ is $\delta = \int_0^1 \frac{d}{ds} \arg \Gamma(\frac{1}{2} + is) ds = \int_0^1 \text{Im} \psi(\frac{1}{2} + is) ds$, where $\text{Im} \psi(1/2 + is) = (\pi/2) \tanh(\pi s)$ is smooth on $[0, 1]$ (no singularity at $s = 0$).

The only input is the standard parameterisation $s = c - \frac{1}{2}$; the value $\Delta s = 1$ follows by arithmetic.

Step 3 (The overlap integral is real). The conical function $P_{-1/2+im}(\cosh \rho)$ is real for all real m and $\rho > 0$ (it is the Laplace integral $(1/\pi) \int_0^\pi (\cosh \rho + \sinh \rho \cos t)^{-1/2} \cos(m \log(\cosh \rho + \sinh \rho \cos t)) dt$, a real integrand). The Yukawa overlap $I(m) = \int_0^{\rho^*} P_{-1/2+im}(\cosh \rho) w(\rho) \sinh \rho d\rho$ (with $w = f_R \varphi_H > 0$) is therefore real and positive ($I(1) = 0.426 > 0$). The positivity $I(m) > 0$ follows analytically: the conical function satisfies $P_{-1/2+im}(\cosh \rho) > 0$ for all $\rho > 0$ and $m \in \mathbb{R}$, by the integral representation $P_\nu(x) = (2/\pi) \int_0^{\pi/2} (x + \sqrt{x^2 - 1} \cos \theta)^\nu d\theta > 0$ for $x > 1$; since $w(\rho) > 0$ and $\sinh \rho > 0$ on $(0, \rho^*)$, every factor in the integrand of $I(m)$ is positive. The normalisation $A = 1/\Gamma(\frac{1}{2} + im)$ is the Harish-Chandra c -function of the spectral decomposition on $\mathbf{H}^2$: it is the unique normalisation for which the conical function satisfies $P_{-1/2+im}(1) = 1$ at the polygon centre (Hejhal 1976, Ch. IV; Harish-Chandra 1958). The phase is normalisation-independent: since $I(m) > 0$ (Step 3), any real rescaling preserves $\arg(Y) = \arg(A) = \arg(1/\Gamma(\frac{1}{2} + im))$. The **entire phase** of the Yukawa coupling $Y = A \times I$ comes from A :

$$\delta = \arg \Gamma(\frac{1}{2} + i) = \int_0^1 \text{Im} \psi(\frac{1}{2} + it) dt = \frac{1}{2} \log \cosh(\pi), \quad (12.9)$$

using $\text{Im} \psi(\frac{1}{2} + it) = (\pi/2) \tanh(\pi t)$ —the same digamma function that generates the Havelock eigenvalue kernel (Paper I, Theorem 4.2). The phase is rephasing-invariant and contributes directly to the Jarlskog invariant.

Step 4 (Evaluation). $\delta = \frac{1}{2} \log \cosh(\pi) = \frac{1}{2} \log(11.592) = 1.225 \text{ rad} = 70.2^\circ$. $\square$

Why the perturbative CKM gives 15° . The tree-level Yukawa diagonalisation yields $\delta_{\text{pert}} = 15.4^\circ$ because it includes only the $w = 0$ and $w = 1$ instanton sectors. Since the

instanton fugacity $\mathcal{K} = 0.548$ (5.2) is $O(1)$, the perturbative expansion in $\mathcal{K}$ does not converge. The scattering phase formula (12.9) resums all sectors exactly via $\arg \Gamma$. The rank-1 't Hooft vertex (the leading non-perturbative correction) moves δ from 15° to 42° at natural strength—confirming the instanton mechanism.

Jarlskog invariant (consistency check). Using the predicted $\delta = 70.2^\circ$ with the *observed* mixing angles gives $J = s_{12}s_{23}s_{13}c_{12}c_{23}c_{13}^2 \sin(70.2^\circ) = 3.12 \times 10^{-5}$ (observed: 3.0×10^{-5} , 4% match). This is a *consistency check*, not an independent prediction: only δ is predicted; the mixing angles are experimental inputs. The CP violation is near-maximal: $\sin \delta = 0.94$. The model's own perturbative CKM matrix gives $J = 3.5 \times 10^{-5}$ (observed: 3.0×10^{-5} , 17% match), which *is* a genuine prediction since all entries are computed from $N = 7$.

12.8 Non-perturbative CKM [non-perturbative]

The non-perturbative CKM combines the BF-crossing phase $\delta = 70.2^\circ$ (Theorem 12.1) with the orbifold image suppression $|V_{cb}| = 0.044$ (§7.3). The CS instanton drives δ from 15° (perturbative) to 70.2° (Theorem 12.1). The instanton simultaneously modifies the mixing angles; at the physical instanton strength ($\alpha \approx 2$, Remark 12.1 below), the pre-orbifold CKM has $|V_{cb}| = 0.067$. The orbifold image suppression (below), with the self-consistent correction to ρ^* , reduces this to:

$$|V_{cb}|_{\text{np}} = 0.044 \pm 12\% \quad (\text{observed: } 0.041 \pm 0.001). \quad (12.10)$$

Orbifold image suppression of V_{cb} . On $\mathbf{H}^2/\mathbb{Z}_7$, the Yukawa coupling for the $c \rightarrow b$ transition receives a suppression from the orbifold images. The nearest $\mathbb{Z}_7$ image of the BF-crossing mode (generation 3, $c = \frac{3}{2}$) sits at geodesic distance

$$d = \text{acosh}(\cosh^2 \rho^* - \sinh^2 \rho^* \cos(2\pi/7)) = 2.02, \quad (12.11)$$

with $\rho^* = 1.734$ (the BO potential zero, Paper V). The Dirac Green's function at this distance is the Legendre Q function of order $\nu = c - \frac{1}{2} = 1$:

$$T_2 = Q_1(\cosh d) = \cosh(d) Q_0(\cosh d) - 1 = 0.024. \quad (12.12)$$

The Yukawa coupling on the orbifold separates into *generation-diagonal* and *generation-changing* channels via the $\text{PSL}(2, \mathbb{F}_7)$ decomposition $\chi_3 \otimes \bar{\chi}_3 = \chi_1 \oplus \chi_8$ (§7.3):

- The χ_1 (singlet) channel gives the diagonal CKM entries ($V_{tb} \approx 1$): the direct Yukawa overlap $T_1 = |V_{cb}|_{\text{pert}} = 0.092$.
- The χ_8 (adjoint) channel gives the off-diagonal entries: the transition $c \rightarrow b$ requires propagation to the nearest $\mathbb{Z}_7$ orbifold image, suppressed by the Dirac Green's function at the image distance d : $T_2 = Q_1(\cosh d) = 0.024$.

The CKM element V_{cb} is the χ_8 -channel transition amplitude between the Galois orbits O_+ and O_- . On the orbifold, the Dirac Green's function decomposes over $\mathbb{Z}_7$ images: $G_{\text{orb}}(x, y) = (1/N) \sum_n G_{\mathbf{H}^2}(x, \gamma^n y)$. The χ_1 (singlet) projection gives the diagonal CKM entries. The χ_8 (adjoint) projection selects the $n \geq 1$ images: $V_{cb} = \langle c | P_{\chi_8} | b \rangle = \sum_{n=1}^6 \langle c | \gamma^n \rangle \langle \gamma^n | b \rangle$. In the spectral representation, the BF-crossing mode ($\lambda_{m^*} \rightarrow 0$, the lightest excitation) dominates. The Green's function at the $n = 1$ image separates into

an *on-site* factor $|\psi_{m^*}(x_c)|^2$ (the mode density at the c -quark position, proportional to the tree-level mixing $|V_{cb}|_{\text{pert}}$) and a *propagation* factor $Q_1(\cosh d)$ (the Dirac propagator at the image distance). Therefore $|V_{cb}|^2 = T_1 \times T_2$ where $T_1 = |V_{cb}|_{\text{pert}}$ (on-site) and $T_2 = Q_1(\cosh d)$ (propagation). The $n = 2$ image at $d_2 = 2d_1 = 4.04$ contributes $Q_1(\cosh d_2)/Q_1(\cosh d_1) = 0.017$ (2% correction). The full series $\sum_{n=1}^{(N-1)/2} Q_1(\cosh(n d_1))$ converges geometrically (ratio $Q_1(\cosh 2d)/Q_1(\cosh d) < 0.02$); the $n \geq 2$ images shift $|V_{cb}|$ by a total of 2.4%, well within the stated $\pm 12\%$ sensitivity.

$$|V_{cb}|^2 = |V_{cb}|_{\text{pert}} \times T_2(\rho^*), \quad |V_{cb}| = \sqrt{0.092 \times T_2}. \quad (12.13)$$

At the uncorrected $\rho_0^* = 1.734$: $|V_{cb}| = 0.047$ (12% off). The orbifold images also shift ρ^* : the BO potential on $\mathbf{H}^2/\mathbb{Z}_7$ is $V_{\text{orb}}(\rho) = V(\rho) + 2 Q_1(\cosh d_1(\rho))$ (the two nearest images at $n = 1$ and $n = 6$, at equal geodesic distances $d_1(\rho)$, each contributing Q_1). Setting $V_{\text{orb}}(\rho^*) = 0$ determines the corrected equilibrium radius. The images lower the BO barrier (same-sign vortex repulsion stabilises the ring), so $V_{\text{orb}}(\rho_0^*) = V(\rho_0^*) - 2Q_1 < 0$: the threshold shifts outward. Linearising $V_{\text{orb}}(\rho_{\text{new}}^*) = 0$ gives

$$\rho^* = \rho_0^* + \frac{2 Q_1(\cosh d_1(\rho^*))}{\coth(\rho_0^*)}. \quad (12.14)$$

Iterating from $\rho_0^* = 1.734$ converges in three steps to $\rho^* = 1.77$, giving $T_2 = Q_1(\cosh d_1) = 0.021$ and

$$|V_{cb}| = \sqrt{0.092 \times 0.021} = 0.044 \pm 12\% \quad (\text{observed: } 0.041 \pm 0.001). \quad (12.15)$$

No new parameters are introduced: the correction is fixed by N and the BO potential. *Sensitivity* (computed numerically): $d \ln |V_{cb}| / d \ln \rho^* = -2.5$, so a 5% shift in ρ^* changes $|V_{cb}|$ by 12%. The precision of the match (6% off) is within the geometric precision tier (factor ~ 2) and rests on ρ^* being the unique zero of the BO potential (determined by N , no free parameters). This suppression applies only to transitions *to* generation 3 (the unique BF-crossing mode, $c: \frac{1}{2} \rightarrow \frac{3}{2}$ crossing $c = 1$). Generations 1 and 2 do not cross the BF threshold ($c > 1$ on both sides), so $|V_{us}| \approx |V_{us}|_{\text{pert}} = 0.237$ (no image suppression).

Wolfenstein parameters from the non-perturbative CKM. All four Wolfenstein parameters are computed from non-perturbative values: $\lambda = |V_{us}| = 0.237$, $A = |V_{cb}|/\lambda^2 = 0.044/0.237^2 = 0.78$, $\gamma = \delta = 70.2^\circ$, and $|V_{ub}|/|V_{cb}| = 0.068$ (from the Yukawa texture). The ratio $|V_{ub}|/|V_{cb}|$ is determined by the $\mathbb{Z}_7$ charge-conserving Yukawa matrix and the $\mathbf{H}^2$ radial profile overlaps: the orbifold image suppression (12.13) changes the *magnitude* of V_{cb} (and V_{ub}) via the BF barrier T_2 , but the unitarity triangle *shape* is set by the Yukawa texture and is invariant under the overall image suppression. With $|V_{ub}| = 0.068 \times 0.042 = 0.003$:

$$\bar{\rho} = \frac{|V_{ub}|}{A\lambda^3} \cos \gamma = 0.098, \quad \bar{\eta} = \frac{|V_{ub}|}{A\lambda^3} \sin \gamma = 0.272.$$

The Jarlskog invariant: $J = A^2 \lambda^6 \bar{\eta} = 2.7 \times 10^{-5}$ (observed: 3.0×10^{-5} , 10% match). $|V_{ub}| = A \lambda^3 \sqrt{\bar{\rho}^2 + \bar{\eta}^2} = 0.003$ (observed: 0.004, 17% off—the weakest element, from the Yukawa texture ratio).

Remark 12.1 (CS instanton and CP redistribution). *The rank-1 't Hooft instanton vertex (fugacity $\mathcal{K} = 0.548$) modifies the perturbative Yukawa matrices. As the instanton*

strength α increases from 0 to 2.0, the CKM phase δ increases from 15° to 70° , providing independent confirmation that the CS instanton drives δ toward the 70.2° derived from the eta invariant (Theorem 12.1). The value $\alpha = 2$ is the minimal instanton action (one unit of Euclidean tunneling).

α	δ	$ V_{cb} $	$ V_{ub} $	V_{ub}/V_{cb}	J
0	15°	0.092	0.006	0.068	3.5×10^{-5}
1.0	42°	0.077	0.005	0.059	5.7×10^{-5}
2.0	70°	0.067	0.006	0.090	1.0×10^{-4}

The instanton redistributes CP violation: the perturbative CKM ($\alpha = 0$) has large $|V_{cb}|$ and small $\sin \delta$, while the non-perturbative CKM has small $|V_{cb}|$ and large $\sin \delta$. The Jarlskog invariant J is not conserved by the instanton—it increases from 3.5×10^{-5} to 1.0×10^{-4} as α increases from 0 to 2. The $|V_{cb}|$ derivation (§7.3) does not use J ; it uses the spectral decomposition of the orbifold Green’s function.

The non-perturbative CKM matrix:

$$|V|_{\text{np}} \approx \begin{pmatrix} 0.972 & 0.237 & 0.003 \\ 0.236 & 0.971 & 0.042 \\ 0.010 & 0.041 & 0.999 \end{pmatrix} \quad \text{Observed: } \begin{pmatrix} 0.974 & 0.224 & 0.004 \\ 0.224 & 0.974 & 0.042 \\ 0.008 & 0.039 & 0.999 \end{pmatrix}$$

13 Consistency checks

13.1 Gravitational consistency

Newton constant. The hierarchy formula (Theorem 5.3) gives $M_P = v \times e^{38.444} = 1.223 \times 10^{19} \text{ GeV}$ (observed: $1.221 \times 10^{19} \text{ GeV}$, 0.15% match). In SI units: $G_N = \hbar c / M_P^2 = 6.65 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ (observed: 6.674×10^{-11} , 0.33% match).

Proposition 13.1 (4D graviton propagator). *The long-range 4D graviton propagator is the standard linearised GR result:*

$$\langle h_{ij}^{(4)}(k) h_{kl}^{(4)}(-k) \rangle = \frac{16\pi G_4}{k_4^2} \mathcal{P}_{ijkl}^{\text{TT}}, \quad (13.1)$$

with 2 tensor polarisations (+, $\times$) and no scalar or vector admixture at $k \ll M_{\text{poly}}$.

Proof. Four steps, each using a standard result.

Step 1 (Boundary $\langle TT \rangle$). In the orbifold CFT at central charge c , the stress tensor 2-point function is fixed by conformal invariance: $\langle T_{ij}(k) T_{kl}(-k) \rangle = (c/24\pi) |k|^3 \mathcal{P}_{ijkl}^{\text{TT}}$, where $\mathcal{P}^{\text{TT}}$ is the transverse-traceless projector. This follows from conservation ($k^i \langle T_{ij} \rangle = 0$), tracelessness ($\delta^{ij} \langle T_{ij} \rangle = 0$), and conformal dimension 2.

Step 2 (Brown–Henneaux map). The normalizable bulk metric perturbation in AdS_3 satisfies $h_{ij}^{(2)}(k) = (16\pi G_3/c) \langle T_{ij}(k) \rangle$ [Brown and Henneaux, 1986]. This is a *tensor* equation: h inherits the $\mathcal{P}^{\text{TT}}$ structure of T .

Step 3 (KK mode sum). The boundary $\mathbb{R} \times S_\theta^1$ lifts on S_φ^1 (circumference $2\pi R = 2\pi/M_{\text{poly}}$) to $\mathbb{R} \times T^2$. The KK mode sum $G(k_3, x_\varphi) = \sum_n G_{2D}(k_3, m_n = n/R) e^{inx_\varphi/R}$ converts the 2D propagator to a 4D one: in the regime $k \gg 1/R$, the sum yields $G^{(4D)} = G_3/(k_4^2 \times 2\pi R) = G_4/k_4^2$. The $\mathcal{P}^{\text{TT}}$ projector is *preserved*: S_φ^1 is orthogonal to the TT projection directions (which act on the $\mathbf{H}^2$ spatial coordinates).

Step 4 (Assembly). Combining Steps 1–3: $\langle h_{ij}^{(4)} h_{kl}^{(4)} \rangle = (16\pi G_4/k_4^2) \mathcal{P}_{ijkl}^{\text{TT}}$, with $G_4 = G_3/(2\pi R) = 1/M_P^2$ (matching the hierarchy formula). In position space, the standard Fourier transform $\int \frac{d^3 k}{(2\pi)^3} e^{i\mathbf{k}\cdot\mathbf{r}}/k^2 = 1/(4\pi r)$ yields Newton’s $1/r$ potential (Corollary 13.2). $\square$

The massive KK modes (graviphoton, radion) contribute additional DOF at $k > M_{\text{poly}}$ but are Yukawa-suppressed ($e^{-r/R} \approx 0$ at $R \sim 10^{-21}$ m) at macroscopic distances. Gravitational radiation at LIGO has tensor polarisation (+, $\times$), matching observations.

Corollary 13.2 (Newton’s law from the Seifert geometry). *The static gravitational potential between two masses m_1, m_2 at separation r on the Seifert spacetime is*

$$V(r) = -\frac{G_N m_1 m_2}{r} \quad (R \ll r \ll \ell), \quad (13.2)$$

with $G_N = G_4 = G_3/(2\pi R)$ and $R = 1/M_{\text{poly}}$. Three distance regimes exist:

1. $r \ll R \approx 10^{-21}$ m: the S^1 fibre is resolved, the geometry is effectively 2+1D, and the dimensionally reduced 3D Newton potential gives $V(r) \sim -(G_3/2\pi) m_1 m_2 \log(r/R)$;
2. $R \ll r \ll \ell$: the KK mode sum yields the standard $1/r$ Newton potential (5.3);
3. $r \gg \ell$: a logarithmic correction $\delta V \sim -(G_3/2\pi) \log(r/\ell)$ from the CS boundary term, with $\delta V/V \sim R/r \times \log(r/\ell) \sim 10^{-19}$ at $r = 1$ m—unmeasurable.

Proof. Tree-level graviton exchange between static sources, using the propagator (13.1), gives the standard result $V(r) = -G_N m_1 m_2 / r$ (see, e.g., Zee *QFT in a Nutshell*, §VIII.1): the Fourier kernel $\int \frac{d^3 k}{(2\pi)^3} e^{i\mathbf{k}\cdot\mathbf{r}}/k^2 = 1/(4\pi r)$, combined with the TT-projector contraction against $T_1^{00} T_2^{00}$, yields (5.3) with $G_N = G_4$. The short-distance regime ($r \ll R$) follows because at distances below the S^1 circumference, the KK modes are unresolved and the zero mode (topological CS sector with no local propagating DOF) dominates, giving 2+1D logarithmic gravity. The long-distance correction ($r \gg \ell$) arises from the CS boundary term, which contributes a non-local logarithmic potential suppressed by $G_3/G_4 = 2\pi R \approx 6 \times 10^{-21}$ m. $\square$

Derivation 13.3 (Full nonlinear 4D Einstein equations).

(Vacuum) *The full nonlinear 4D vacuum Einstein equations $R_{MN}^{(4)} = \Lambda^{(4)} g_{MN}$ on $\mathbb{R} \times (\mathbf{H}^2 \times_N S^1)$ follow from the $\text{SL}(2, \mathbb{R})$ CS theory (Steps A–B below).*

(Matter) *The matter-coupled equations $G_{MN} + \Lambda g_{MN} = 8\pi G T_{MN}$ follow from the Lashkari entanglement argument (Step C below), using the Ryu–Takayanagi formula (derived in 2+1D from Cardy–Calabrese + Brown–Henneaux, Step C).*

Proof. All three steps are self-contained in 2+1D.

Step A (CS $\rightarrow$ 3D nonlinear Einstein). The CS flatness condition $F = dA + A \wedge A = 0$ (with $A = \omega + e/\ell$) is equivalent to the 3D vacuum Einstein equation $R_{ij}^{(3)} = \Lambda_3 g_{ij}$ plus the torsion-free condition $T^a = 0$ [Witten, 1988]. This is exact: the full nonlinear equations, not a linearisation.

Step B (KK lift to 4D). On the Seifert manifold with stabilised fibre radius ($\sigma = 1$, from the flux potential $N/2$) and quantised flux ($F = N/(2\pi)$, from the Chern class), the 4D Ricci tensor decomposes as: $R_{ij}^{(4)} = R_{ij}^{(3)} - \frac{1}{2} F_{ik} F_j{}^k$, $R_{i\varphi}^{(4)} = \frac{1}{2} \nabla_j F^j{}_i = 0$ (uniform

flux satisfies Maxwell on constant-curvature $\mathbf{H}^2$), $R_{\varphi\varphi}^{(4)} = -\frac{1}{4}F^2$ (constant for uniform flux). The $\varphi\varphi$ component gives $\Lambda^{(4)} = N^2/16$ (in units of $1/\ell^2$). The ij components give $R_{ij}^{(3)} = \Lambda_3 g_{ij}$ with $\Lambda_3 = (N^2 - 16)/16$ —precisely the 3D equation that the CS theory produces in Step A. The mixed component $R_{i\varphi}^{(4)} = 0$ is automatically satisfied. Therefore the full 4D vacuum Einstein equations hold on the Seifert background. For general perturbations around this background, the linearised equations follow from the graviton propagator (Proposition 13.1); the nonlinear matter-coupled equations follow from the entanglement argument (Step C below).

Step C (Matter coupling from entanglement — RT derived). The vacuum equations (Steps A–B) are derived without AdS/CFT. The matter coupling uses the Lashkari et al. [2014] entanglement argument, which requires the Ryu–Takayanagi formula. In 2+1D, RT is derived from the boundary CFT:

1. The Cardy–Calabrese formula $S_{\text{EE}} = (c/3)\log(L/\varepsilon)$ for a boundary interval of length L is a *theorem* of CFT_2 (conformal Ward identity for the twist-field correlator, no gravity input);
2. the Brown–Henneaux relation $c = 3\ell/(2G)$ (our consistency check, §5);
3. the geodesic length on $\mathbf{H}^2$: $\text{Length}(\gamma) = 2\ell\log(L/\varepsilon)$ (Riemannian geometry, no dynamics).

Combining: $S_{\text{EE}} = (c/3)\log(L/\varepsilon) = \ell\log(L/\varepsilon)/(2G) = \text{Length}(\gamma)/(4G)$ — the RT formula, derived from CFT_2 and geometry. The Lashkari argument then uses three ingredients: (i) the boundary CFT is *unitary*: the $\mathbb{Z}_N$ orbifold of a unitary CFT by a finite group is unitary [Dixon et al., 1985], and the starting theory (the free boson on S^1 at each angular mode, with total $c = 12b(N)$) is unitary; (ii) RT holds (derived above); (iii) relative entropy is non-negative ($S(\rho\|\sigma) \geq 0$, a theorem of quantum information). The positivity of relative entropy for all ball-shaped boundary regions and all states implies $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G \langle T_{\mu\nu} \rangle$ —the full nonlinear Einstein equations with matter—with $G = 3/(2c)$ from Brown–Henneaux. No additional thermodynamic argument (Jacobson, Clausius) is needed. The vacuum equations are exact (Steps A–B); the matter coupling is established at *linearised* order (first order in $T_{\mu\nu}$) by Lashkari et al. [2014]. The extension to second order uses the quantum-corrected RT formula [Faulkner et al., 2014]; in 2+1D this is derived (the Cardy–Calabrese formula includes all quantum corrections at finite c), so the second-order extension is also self-contained. The 3D matter coupling lifts to 4D by the same KK reduction as Step B (stabilised fibre, quantised flux). $\square$

Equivalence principle. Different fermion generations carry different KK charges m , so the *graviphoton* force is composition-dependent. However, this force has Yukawa range $R \sim 10^{-21}$ m (10^5 times smaller than the proton). At macroscopic distances, only universal forces survive: the topological CS sector (which couples to topology, not KK charge) and the radion (which couples to the trace of $T_{\mu\nu}$, universally). The Eötvös parameter $\delta(a)/a \lesssim e^{-r/R} \lesssim e^{-10^{20}} \approx 0$, easily satisfying the MICROSCOPE bound $\delta(a)/a < 10^{-15}$.

13.2 Gauge-theory consistency

Anomaly cancellation. Each generation carries the standard $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ quantum numbers (left-handed Weyl convention). All gauge anomalies ($\text{Tr}(Y)$, $\text{Tr}(Y^3)$,

$\text{Tr}(T_3^2 Y))$ cancel per generation by the same arithmetic as in the Standard Model; the orbifold determines which modes form a generation, not what quantum numbers they carry.

Proton stability. The $\text{SU}(3)$ and $\text{SU}(2)$ sectors live in different polygon numbers ($N = 7$ and $N = 4$). Cross-sector Yukawa couplings require $8(m_1 + m_2) \equiv 28 \pmod{56}$, which has no integer solution ($\gcd(8, 56) = 8$ does not divide 28). This is exact $\mathbb{Z}_{56}$ charge conservation. *Prediction:* the proton is exactly stable—not just long-lived—distinguishing this theory from $\text{SU}(5)$ GUTs ($\tau_p \sim 10^{34}$ yr) and consistent with Super-Kamiokande bounds ($\tau_p > 1.6 \times 10^{34}$ yr for $p \rightarrow e^+ \pi^0$).

13.3 The Higgs mass

The electroweak VEV $v = 246$ GeV is predicted (Theorem 5.3, 0.12% match). Standard Model RG running of λ_H from M_{poly} to M_Z gives $\lambda_H \approx 0.10$, yielding $m_H \approx 110$ GeV at tree level. Two corrections, both derived from the orbifold structure, raise this:

1. *Twisted-sector quartic.* The $\mathbb{Z}_7$ orbifold has $N - 1 = 6$ twisted sectors. Each twisted-sector operator is localised at the orbifold fixed point with OPE coefficient $C_{\text{tw}} \propto 1/\sqrt{N}$ (the localisation suppression from DHVW [Dixon et al., 1985]). The four-point function receives a correction from all twisted sectors: $N \times |C_{\text{tw}}|^4 = N \times 1/N^2 = 1/N$. The charge selection rule ($2m_3 + 2m_4 = 14 \equiv 0 \pmod{7}$) restricts the quartic to $|\varphi_3|^2 |\varphi_4|^2$ (the unique $\mathbb{Z}_7$ -invariant quartic). Therefore: $\delta\lambda_{\text{tw}} = \lambda_{\text{SM}}/N = 0.10/7 = 0.014$.
2. *Instanton quartic.* The physical CS level $k_{\text{phys}} = c/6 - 1/2 = 8.096$ (at $N = 7$) has fractional part $\kappa = 0.096$. The instanton fugacity is $\mathcal{K} = 0.548$ (5.2). The quartic vertex involves two instanton–anti-instanton pairs (amplitude $\mathcal{K}^2 = 0.300$), mediated by the critical mode ($j = 1$, conformal weight $h = j(j+1)/(k+h^\vee) = 2/3$). The critical mode dominates because it is the lightest excitation at the BF threshold ($\lambda_{m^*} = 0$). The central-charge suppression $1/c_{\text{EW}} = 1/17.2$ (the inverse gravitational coupling in the EW sector) gives: $\delta\lambda_{\text{inst}} = \mathcal{K}^2 h^2 / c_{\text{EW}} = 0.300 \times 0.444 / 17.2 = 0.008$.

At leading order: $\lambda_H^{\text{LO}} = 0.10 + 0.014 + 0.008 = 0.122$, giving $m_H^{\text{LO}} = \sqrt{2\lambda_H} v = 122$ GeV. Four next-to-leading corrections raise this:

3. $O(1/N^2)$ twisted sector: $\delta\lambda = \lambda/N^2 = 0.002$ (subleading DHVW OPE coefficient).
4. $O(\mathcal{K}^4)$ instanton: $\delta\lambda = 6 \times 10^{-5}$ (negligible).
5. *Two-loop SM running:* $\delta\lambda = +0.007$ (the standard top-quark two-loop correction $3y_t^4/(16\pi^2)^2 \times [16g_s^2 - 9y_t^2/2] \times \ln(M_{\text{poly}}/m_t)$).
6. *Top threshold:* $\delta\lambda = +0.004$ (finite EW-scale matching).

At next-to-leading order: $\lambda_H^{\text{NLO}} = 0.122 + 0.013 = 0.135$, giving

$$m_H = \sqrt{2\lambda_H^{\text{NLO}}} v = 128 \text{ GeV}. \quad (13.3)$$

The observed $m_H = 125.25$ GeV lies between the LO (122) and NLO (128) values; the theoretical uncertainty from $O(1/N^3)$ and three-loop corrections is ± 3 GeV. Every coefficient ($1/N$, κ , h , c_{EW} , y_t , g_s) is determined by $N = 7$ and the SM particle content.

13.4 Predictions and cross-checks

Black hole entropy. The Cardy formula of the orbifold CFT at $c = 12b(N)$ gives

$$S_{\text{BH}} = 2\pi\sqrt{\frac{c}{6}E_L} + 2\pi\sqrt{\frac{c}{6}E_R}, \quad (13.4)$$

which reproduces the Bekenstein–Hawking entropy $S = \pi r_+/(2G_3)$ via the Brown–Henneaux relation $c = 3\ell/(2G_3)$. This is the standard Strominger (1998) microstate counting for the BTZ black hole. The microstates are Virasoro descendants of the orbifold CFT; the Havelock central charge $c = 51.57$ gives $S = 18.4\sqrt{E}$ with sub-leading correction $-\frac{3}{2}\log E + \dots$

The polygon $\rightarrow$ BTZ transition is a *Hawking–Page transition*: at each palindromic threshold N , the entropy jumps by $\Delta S = \log 2$ per pair (wall-crossing). The $\mathbb{Z}/2\mathbb{Z}$ palindromic symmetry ($m \leftrightarrow N-m$) gives 1 bit per self-paired mode (N even) and 2 bits per paired mode (N odd).

Under the KK lift to 4D, the BTZ entropy becomes the 4D Bekenstein–Hawking entropy via $G_4 = G_3/(2\pi R)$, giving $S_{\text{4D}} = A/(4G_4)$ for the 4D horizon area A .

Code availability. All numerical verifications are in the companion repository (<https://github.com/gspea/planetary-polygons-unified>). Parameter accounting and predictions appear in Paper VI.

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